When Geopolitical Shocks Become Systemic: Volatility Connectedness and Market-Implied Stress in European Equity Markets

[Author information omitted for blind review]

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Abstract

We ask when geopolitical shocks become systemic financial stress events. Using a daily panel of 19 European equity markets over 2017–2025, we construct a European Market Fragility Index (EMFI) from principal component analysis and a Hidden Markov Model that assigns each trading day to latent stress states without reference to geopolitical events. Local projections on 154 detected geopolitical shock days yield positive but statistically imprecise average responses: 63% of shocks arrive when markets are calm and produce negligible fragility effects. State-dependent projections reveal economically large amplification in already-fragile markets. TCI amplification is precisely estimated ($\hat{\theta}_{k=0} = 2.70$, $p = 0.090$; total effect at $P_{\text{stress}} = 0.9$ of 2.69 units vs. 0.09 in calm conditions); EMFI amplification is economically large but statistically imprecise ($\hat{\theta}_{k=0} = 1.44$, $p = 0.279$; 1.43 units vs. 0.14 in calm states). A predictive logit model estimated on pre-shock market conditions achieves a leave-one-out AUC of 0.691: a one-standard-deviation rise in pre-shock EMFI increases the odds of a systemic shock outcome by a factor of 6.3. The event taxonomy classifies 34 of 138 geopolitical shocks as Systemic (24.6%), concentrated in the Ukraine conflict cluster of 2022 and the European equity correction of August 2024. The central finding is that the financial consequences of geopolitical events depend not primarily on the shocks themselves but on the fragility of the markets they enter.

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1 Introduction

Geopolitical conflict is an increasingly prominent feature of the financial landscape. From the Russian invasion of Ukraine in February 2022 to the Hamas–Israel conflict of October 2023, geopolitical events periodically disrupt global equity markets with sudden and severe consequences. Yet the vast majority of geopolitical news events—conflicts, territorial disputes, diplomatic crises—leave no discernible trace in aggregate market fragility. The central question of this paper is: *when do geopolitical shocks become systemic financial stress events?*

Existing research on geopolitical risk and financial markets has focused primarily on average return or volatility responses to conflict-related news [10, 12, 15]. These studies demonstrate that geopolitical uncertainty raises risk premia and depresses equity valuations on average, but they do not address the transmission to *systemic* fragility—the co-movement, tail co-exceedance, and volatility connectedness that characterise genuine crises. A shock that raises the volatility of a single market is qualitatively different from one that simultaneously pushes 18 out of 19 European equity markets into their return tails, elevates cross-market connectedness above 90%, and causes the market-implied probability of systemic stress to reach unity. Explaining the difference between these two types of shock response is the contribution of this paper.

This paper makes three contributions.

First, we construct a market-based systemic fragility framework that is explicitly independent of the geopolitical shock series. The framework combines a European Market Fragility Index (EMFI)—a principal-component composite of volatility stress, tail co-exceedance breadth, rolling correlation, and total volatility connectedness (TCI, Diebold and Yilmaz 13)—with a Hidden Markov Model that assigns each trading day to one of three latent stress states (calm, elevated, systemic) without reference to geopolitical event dates. The geopolitical shock trigger is taken from the validated high-frequency series of Lupu et al. [23], which applies extreme value theory and false-discovery-rate testing to GDELT Document 2.0 conflict coverage for 19 European countries.

Second, we provide evidence on state-dependent geopolitical shock transmission. Unconditional local projections yield positive but imprecise average responses—the expected result when 63% of shocks strike calm markets that produce negligible fragility effects. Smooth-transition local projections, conditioning on the lagged HMM stress probability, reveal that shocks in already-fragile markets generate substantially larger fragility responses. Under near-systemic conditions ($P_{\text{stress},t-1} = 0.9$), the TCI total effect is 2.69 percentage points vs. 0.09 in calm markets ($\hat{\theta}_{k=0} = 2.70$, $p = 0.090$); the EMFI total effect is 1.43 standard deviations vs. 0.14 ($\hat{\theta}_{k=0} = 1.44$, $p = 0.279$). Amplification ratios of approximately 10× obtain for both outcomes, though we caution that ratios are sensitive to the small calm-market baseline.

Third, we develop a predictive taxonomy of geopolitical shock outcomes. A binary logit estimated purely on pre-shock market conditions achieves a leave-one-out AUC of 0.691, with pre-shock EMFI as the dominant predictor (odds ratio 6.3). This establishes that the systemic/absorbed distinction is not post-hoc labelling but reflects a genuine predictable difference in market states at the time of the shock.

With this apparatus in hand, we estimate Jordà [2005] local projections (LPs) to trace the dynamic response of each fragility outcome to a geopolitical shock. We further test for state dependence: do shocks produce larger fragility responses when they strike already-fragile markets? Finally, we classify the 138 geopolitical shock events into an *absorbed / localised / systemic / market-only* taxonomy based on the joint behaviour of the EMFI and $P_{\text{stress},t}$ in the days following each event.

The results are organised around three findings.

First, the unconditional geopolitical shock effect is positive but statistically imprecise—a motivated null rather than a limitation. The EMFI response at the impact horizon is $\hat{\beta}_0 = 0.245$ (COVID-zeroed primary baseline), not significantly different from zero ($p = 0.624$). This is the expected outcome when 63% of shocks arrive in calm market conditions: the large responses in fragile states are averaged with near-zero responses in the absorbed majority.

Second, state-dependent projections reveal that the shock-fragility relationship is highly nonlinear. Under near-systemic pre-existing fragility ($P_{\text{stress},t-1} = 0.9$), a geopolitical shock raises TCI by 2.69 percentage points (SE = 1.59) compared with 0.09 in calm conditions ($\hat{\theta}_{k=0} = 2.70$, $p = 0.090$; block-bootstrap 95% CI excludes zero). For EMFI, the near-systemic effect is 1.43 standard deviations (SE = 1.27) versus 0.14 in calm conditions ($\hat{\theta}_{k=0} = 1.44$, $p = 0.279$)— economically large but statistically imprecise. These level differences correspond to amplification ratios of approximately 10× for both outcomes; we caution that such ratios are sensitive to the small calm-market baseline. The TCI amplification is robust across 12 alternative specifications.

Third, the predictive taxonomy confirms that the systemic/absorbed distinction reflects pre-shock market conditions, not post-hoc classification. The logit model's leave-one-out AUC of 0.691 establishes that pre-shock fragility carries genuine out-of-sample predictive content: a one-SD rise in EMFI_{t-1} increases the odds of a systemic outcome by a factor of 6.3. The Hamas–Israel attack of October 2023 is classified Systemic via a one-day delayed stress response (market-implied probability reaches 1.0 the trading day after the attack), illustrating that the post-event evaluation window matters for accurate classification.

The paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the data and shock series. Section 4 constructs the fragility measures and EMFI. Section 5 presents the HMM stress-state model. Section 6 reports the baseline LP results. Section 7 presents state-dependent LPs, the event taxonomy, and the predictive classification model. Section 8 discusses robustness checks. Section 9 concludes.

2 Related Literature

Geopolitical risk and financial markets. A large literature documents that geopolitical uncertainty affects equity valuations, risk premia, and trading activity. Caldara and Iacoviello [12] construct a monthly geopolitical risk (GPR) index from newspaper coverage and show that elevated GPR reduces investment and employment. Berkman et al. [10] find that wars and military conflicts are associated with positive equity returns upon resolution, consistent with a risk-premium channel. Liu and Zhang [21] demonstrate that geopolitical risk is priced in the cross-section of equity returns, while Pástor and Veronesi [25] show that political uncertainty generates time-varying risk premia even absent fundamental cash-flow shocks. Smales [28] documents significant volatility spillovers from geopolitical risk to oil and equity markets, with transmission amplified during acute conflict episodes; Antonakakis et al. [6] find geopolitical risk is a structural driver of the oil–stock covariance over the long run. Su et al. [30] and Jiang and Yu [18] extend this work to cross-market synchronisation and bilateral spillovers using news-based geopolitical risk proxies. Caldara and Iacoviello [12] construct a monthly news-based Geopolitical Risk (GPR) index spanning 1900–2019 and document that elevated geopolitical risk raises macroeconomic uncertainty and depresses equity valuations, yet leave open the question of systemic transmission and fragility-conditioned amplification that motivates the present paper.

Systemic risk and volatility connectedness. Following the 2008 global financial crisis, a substantial literature has developed measures of systemic risk and cross-market contagion. Diebold and Yilmaz [13] introduce a forecast-error variance decomposition approach (the total connectedness index, TCI) based on rolling VAR models, which has become a standard tool for measuring the degree to which idiosyncratic shocks propagate across a system. Acemoglu et al. [1] model how network structure determines whether local shocks cascade or are absorbed. Acharya et al. [2] measure systemic risk through marginal expected shortfall (MES), while Adrian and Brunnermeier [3] develop CoVaR as a measure of each institution's systemic risk contribution. Brownlees and Engle [11] extend the MES framework to SRISK, quantifying the capital shortfall of a financial institution conditional on a severe market decline. Lo Duca and Peltonen [22] develop a macro-financial early-warning framework for predicting systemic events that is explicitly designed for European markets, using composite indicators that share conceptual ground with our EMFI. Our index builds on this tradition by combining connectedness with co-exceedance tail breadth and rolling correlation into a single composite.

European equity market connectedness and international contagion. A growing literature examines how financial stress propagates across European equity markets, which are highly integrated but heterogeneous in fragility. Bekaert et al. [8] document significant contagion across 55 equity markets during the 2007–2009 global financial crisis, with European markets among the most susceptible to cross-border transmission. Diebold and Yilmaz [14] measure trans-Atlantic equity volatility connectedness between U.S. and European financial institutions over 2004–2014 and find that the direction of spillovers shifted from predominantly U.S.-to-Europe during 2007–2008 to bi-directional from 2008 onwards, reflecting the European sovereign debt crisis as a new shock origin. These findings motivate our focus on European equity markets as a test laboratory: markets that are deeply integrated yet exposed to distinct geopolitical shock origins (energy supply disruptions, Eastern-European conflict risk) since 2022.

Local projections. Jordà [19] proposes local projections as a flexible alternative to VAR-based impulse responses: each horizon is estimated in a separate regression, which accommodates non-linearities and is robust to misspecification of the full dynamic system. Romer and Romer [27] and Stock and Watson [29] discuss identification strategies for causal LP inference. Jordà and Taylor [20] and Auerbach and Gorodnichenko [7] develop state-dependent LP specifications. We apply the standard Jordà LP framework with Newey-West standard errors, following the now-standard practice in applied macroeconometrics.

Hidden Markov Models in finance. Hamilton [17] introduces the Markov-switching model to capture regime changes in macroeconomic time series. Guidolin and Timmermann [16] and Ang [4] apply HMMs to asset allocation and risk management. Ang and Bekaert [5] demonstrate that international equity portfolios benefit from regime-switching specifications because bear-market correlations across countries are substantially higher than bull-market correlations—the asymmetry that motivates our HMM treatment of European market fragility states. We use a multivariate Gaussian HMM to classify market fragility states purely from market-based indicators, ensuring that the classification is independent of the geopolitical shock series used in the LP regressions.

3 Data and Geopolitical Shock Triggers

3.1 Equity return panel

Our equity return panel covers 19 European markets over the period 2017-01-02 to 2025-10-15, comprising 2,278 trading days.[1] Daily log returns are constructed from total-return equity market indices sourced from LSEG Workspace. The 19 countries span all major Western and Central European markets. Table 1 lists all countries and their benchmark equity indices.

Table 1: Equity market panel: country coverage

Country	Benchmark index
Austria	ATX
Belgium	BEL 20
Bulgaria	SOFIX
Croatia	CROBEX
Finland	OMX Helsinki
France	CAC 40
Germany	DAX
Greece	ATHEX Composite
Hungary	BUX
Ireland	ISEQ Overall
Italy	FTSE MIB
Netherlands	AEX
Norway	OBX
Poland	WIG20
Portugal	PSI 20
Romania	BET
Spain	IBEX 35
Sweden	OMX Stockholm 30
United Kingdom	FTSE 100

Daily log returns sourced from LSEG Workspace. Study period: 2017-01-02 to 2025-10-15 (2,278 trading days). Cluster assignments (3-group geographic scheme) are reported in Appendix Table 9.

3.2 Geopolitical conflict shock series

We use the validated geopolitical conflict shock series of Lupu et al. [23] without modification.[2] We briefly summarise the series properties relevant to the present application.

The shock series is derived from country-level conflict-related news coverage in the GDELT Document 2.0 database, transformed into daily tail-based shock intensities via generalised Pareto distribution (GPD) peaks-over-threshold methods and Benjamini–Hochberg false discovery rate

[1] Fifteen calendar dates present in the GDELT conflict index are absent from the equity price series and are dropped; the study period therefore begins on the first day on which all 19 return series are available.

[2] The present paper takes the shock series as a given input; shock construction methodology is a contribution of Lupu et al. [23] and is not repeated here.

(FDR) corrections [9]. For each country i and calendar day t, the shock intensity is

$$S_{i,t} = -\log q_{i,t}, \tag{1}$$

where $q_{i,t}$ is the BH-FDR adjusted tail exceedance probability; days with raw tail probability $p_{i,t} < 0.005$ are classified as binary shock events, while the continuous $S_{i,t}$ serves as the LP regressor. Consecutive exceedance events within a 7-day window are declustered to avoid double-counting a single geopolitical episode. Full methodological details are provided in Lupu et al. [23].

The resulting shock series contains 278 declustered shock events over the study period, with shock intensities $S_{i,t}$ ranging from 0.073 to 2.860 (mean 0.318). Of these events, 181 (65.1%) fall on weekends or holidays when equity markets are closed; these are forward-filled to the next available trading day, yielding 163 days on which the aggregate shock measure $\text{MaxShock}_t = \max_i S_{i,t}$ is positive.[3]

The aggregate shock measure used as the LP regressor is MaxShock_t—the maximum shock intensity across all 19 countries on day t. This captures the most severe shock on each day and is interpretable as the single strongest geopolitical signal reaching the market. Alternative aggregation schemes (AvgShock_t, BreadthShock_t) are used in robustness checks (Section 8).

4 Market-Based Financial Stability Measures

4.1 Daily volatility stress and tail co-exceedance

We construct seven daily cross-sectional fragility indicators from the 19-country equity return panel. All indicators are computed from the return series without reference to geopolitical shock dates, ensuring that the measurement of market fragility is independent of the shock variable used in subsequent regressions.

Volatility stress and squared stress. The first two indicators measure the average magnitude of daily moves:

$$\text{VolStress}_t = \frac{1}{N}\sum_{i=1}^{N} |r_{i,t}|, \tag{2}$$

$$\text{SqStress}_t = \frac{1}{N}\sum_{i=1}^{N} r_{i,t}^2, \tag{3}$$

where $N = 19$ is the number of countries. VolStress reached a maximum of 0.119 on 2020-03-16 (the peak of the COVID-19 equity crash), when the cross-sectional average of daily absolute returns across European markets exceeded 11.9%.

[3] The local projections and event taxonomy use 154 of these 163 shock-day observations. Nine shock days fall before 2017-05-19, the start of the EMFI/HMM estimation window (which requires a warm-up period for the rolling TCI and tail-breadth calculations); these nine observations are excluded from the LP sample but remain in the full-sample market panel. Of the remaining 154 shock observations, 16 fall within calendar year 2020 and are identified as COVID-19-related news events (see Section 8.1). Because COVID-19 is a systemic financial shock but not a geopolitical conflict shock, these 16 days are excluded from the LP *treatment* series in the primary specification; the market outcomes (EMFI, TCI, etc.) are retained in the sample for the purposes of EMFI and HMM validation. The primary LP therefore uses 138 geopolitical shock-day observations as treatment events.

Tail co-exceedance breadth. The tail breadth indicators count how many markets simultaneously experience extreme moves on day t:

$$\text{TailVolBreadth}_t = \#\{i : |r_{i,t}| > \hat{q}_{i,0.95,t-1}\}, \tag{4}$$

$$\text{TailLossBreadth}_t = \#\{i : r_{i,t} < \hat{q}_{i,0.05,t-1}\}, \tag{5}$$

where $\hat{q}_{i,p,t-1}$ is the p-th empirical quantile of country i's absolute (or signed) return computed over the 252 trading days prior to day $t-1$.[4] This rolling-threshold design accounts for the country-specific baseline level of volatility and is strictly backward-looking, requiring a minimum of 60 historical observations. On 2020-03-16, TailVolBreadth reached 18 out of 19—virtually the entire cross-section simultaneously exceeded its country-specific 95th percentile of absolute returns, a characterisation of extreme systemic co-movement.

Rolling average pairwise correlation. We compute the average of all $N(N-1)/2 = 171$ pairwise Pearson return correlations over rolling 60-day and 30-day windows:

$$\text{AvgCorr}_{W,t} = \frac{2}{N(N-1)} \sum_{i<j} \hat{\rho}_{ij,t}^{(W)}, \tag{6}$$

where $\hat{\rho}_{ij,t}^{(W)}$ is the rolling Pearson correlation estimated over the W days ending at t. The 60-day measure has a sample mean of 0.476 and ranges from 0.207 to 0.871, reflecting the high degree of structural financial integration within European equity markets.

4.2 Total volatility connectedness index (TCI)

The total connectedness index follows Diebold and Yilmaz [13] and Pesaran and Shin [26]. For each rolling window of $W = 100$ trading days, we estimate a VAR(p) model on the 19-dimensional vector of absolute returns $\mathbf{v}_t = (|r_{1,t}|, \dots, |r_{19,t}|)'$, selecting the lag order $p \in \{1, 2, 3\}$ by BIC. We then compute the generalised forecast-error variance decomposition (GFEVD) [26], which is order-invariant and permits each variance share to reflect both direct and indirect spillover paths. The normalised GFEVD matrix $\tilde{\Theta}$ has rows summing to one by construction; the TCI is the fraction of the total forecast-error variance that is attributable to cross-market spillovers:

$$\text{TCI}_t = \frac{\sum_{i \neq j} \tilde{\theta}_{ij,t}}{N} \times 100. \tag{7}$$

The sample mean of TCI_t (W=100) is 70.5%, reflecting the deep structural integration of European equity markets. The index ranges from 46.7% to 94.6% and spikes above 80% during the COVID-19 crash, confirming that near-total volatility co-movement characterised that episode. BIC selects $p = 1$ in the large majority of rolling windows, consistent with the well-documented predominance of first-order dynamics in high-frequency volatility data.

An important caveat is that the sample average TCI of 70.5% reflects structural financial integration, not acute stress: TCI is always high in integrated markets, and it is elevation *above* its time-series mean that constitutes the relevant signal. We also note that TCI and AvgCorr60 carry a high within-sample correlation (0.797); Section 8 tests whether the two measures are empirically substitutable in the LP regressions.

[4] The one-day lag on the threshold ensures strict out-of-sample construction: the threshold on day t depends only on information available by close of day $t-1$.

A potential concern is overparameterisation: a 19-variable VAR with p lags and a window of W days must estimate $19^2p + 19$ parameters from only W observations per equation. With $W = 100$ and BIC typically selecting $p = 1$, each of the 19 equations has 19 slope coefficients estimated from 100 observations — a ratio that is conservative relative to standard econometric practice. Nonetheless, we directly address this concern via two channels. First, we recompute the TCI using a wider window of $W = 250$ trading days ($\approx$1 calendar year), fixing $p = 1$; this improves the observation-to-parameter ratio fivefold. The resulting TCI series has a sample mean of 71.1%, closely tracking the $W = 100$ series (70.5%), and the window-sensitivity table in Appendix E confirms that panel LP $\hat{\beta}_{k=0}$ and state LP $\hat{\theta}_{k=0}$ are both positive and economically meaningful across all windows $W \in \{60, 100, 150, 200, 250\}$. Second, we supplement the VAR-based TCI with two parameter-free network metrics derived from the rolling correlation matrix: the first eigenvalue scaled by N (λ_1/N), which measures overall co-movement, and the network density (fraction of cross-market correlation pairs exceeding 0.5). Both metrics spike sharply at the COVID crash and the Ukraine invasion, corroborating that the TCI captures genuine stress episodes rather than VAR over-fit artefacts.

4.3 European Market Fragility Index (EMFI)

We combine four of the above indicators into a composite daily index via principal component analysis (PCA). The four components are standardised to mean zero and unit standard deviation using their full-sample means and standard deviations (consistent with the EMFI being an *ex-post* outcome variable, not a real-time detector). Standardised versions are denoted $\tilde{X}_t$ throughout.

PCA is applied to the covariance matrix of

$$\mathbf{X}_t = (\widetilde{\text{VolStress}}_t,\ \text{Tail}\widetilde{\text{VolBreadth}}_t,\ \text{Avg}\widetilde{\text{Corr}}60_t,\ \widetilde{\text{TCI}}_t^{(100)})', \tag{8}$$

and the first principal component is extracted. The sign is fixed so that a higher EMFI corresponds to greater fragility (orientation by sign of the correlation between PC1 and VolStress).

The first principal component explains 58.3% of the joint variance of the four components, above the 50% threshold we adopt as a credibility criterion for a composite. The loadings are all positive and balanced: VolStress (0.532), AvgCorr60 (0.516), TCI (0.495), TailVolBreadth (0.454). Notably, the second principal component accounts for 31.7% of variance, reflecting a two-cluster covariance structure in the input data. Specifically, VolStress and TailVolBreadth are highly correlated (0.784) with each other but only weakly correlated (0.16–0.38) with AvgCorr60 and TCI, which are themselves highly correlated (0.797). The first PC therefore averages the two clusters; the second PC captures the contrast between acute volatility stress and structural co-movement. This two-cluster structure is acknowledged throughout the paper: the EMFI is a balanced composite across both dimensions of fragility rather than a near-perfect common factor.

The EMFI is available from 2017-05-19 (after the warm-up periods of all four components are satisfied) to 2025-10-15, spanning 2,179 trading days. It peaks at 15.22 on 2020-03-12 (COVID-19 crash) and at approximately 3–4 units during the Ukraine invasion episode. The 75th percentile of the full-sample distribution is 0.55, which defines the `HighFragility` state used in Section 7.

Figure 1 plots the EMFI over the study period alongside its four standardised components, with shading for the COVID-19 crash, Ukraine invasion, and Hamas–Israel episodes.

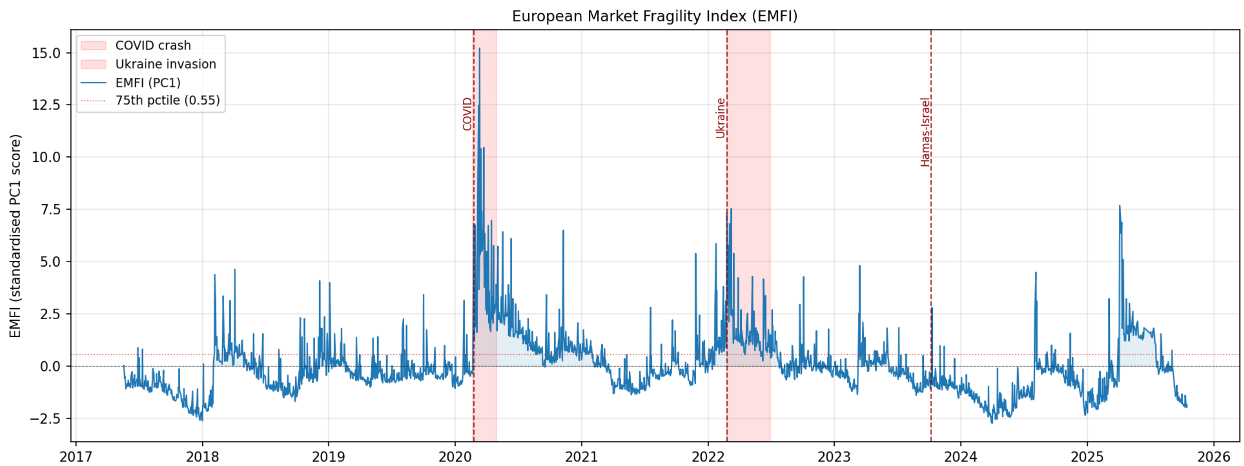

Figure 1: European Market Fragility Index (EMFI) and components, 2017–2025. Shaded bands mark the COVID-19 crash (Feb–Apr 2020), Ukraine invasion (Feb–Mar 2022), and Hamas–Israel attack (Oct 2023).

5 Market-Implied Systemic Stress States

5.1 Model specification

We estimate a multivariate Gaussian Hidden Markov Model (HMM) on the standardised EMFI component vector $\mathbf{X}_t$ defined in Section 4.3. The HMM assumes that the observed $\mathbf{X}_t$ is generated by a latent Markov chain with K states, each associated with a multivariate Gaussian emission distribution:

$$\mathbf{X}_t \mid s_t = k \ \sim \ \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \qquad k = 1, \ldots, K, \tag{9}$$

where s_t denotes the latent state at day t. Parameters are estimated by the Baum-Welch (EM) algorithm with 50 random initialisations; we retain the solution with the highest log-likelihood.

We set $K = 3$ as the primary specification and label the states post-hoc in ascending order of their mean EMFI value: State 0 (Calm), State 1 (Elevated), State 2 (Systemic stress). A two-state model is used as robustness (Section 8).

A critical methodological constraint is that the HMM is estimated entirely from market-based fragility measures and contains no information about geopolitical shock dates. The daily posterior probability of systemic stress, $P_{\text{stress},t} \equiv \Pr(s_t = 2 \mid \mathbf{X}_{1:t})$, is therefore an independent market-implied measure that can be used as a conditioning variable in the LP regressions of Section 7.

5.2 Estimated stress-state structure

The estimated three-state HMM converged to a log-likelihood of -1367.1 over the 2,179-day sample. Table 2 reports the transition matrix and state-specific emission means.

Several features of the estimated model are notable. *First*, calm is the dominant state (63.7% of days), consistent with the prior that European equity markets spend most time in non-crisis conditions. *Second*, the calm state is strongly persistent (diagonal 0.719), while the systemic-stress state is highly transient (diagonal 0.371, implying a mean duration of approximately 1.6 trading days). The systemic state therefore captures episodic spikes rather than prolonged stress episodes—a finding consistent with the acute-stress nature of the EMFI documented in Section 4.3. *Third*, when

Table 2: HMM transition matrix and emission statistics

	Transition probabilities			Viterbi path	
From \ To	Calm	Elevated	Systemic	Days	Share
Calm (0)	0.719	0.212	0.069	1,389	63.7%
Elevated (1)	0.526	0.377	0.097	567	26.0%
Systemic (2)	0.411	0.217	0.371	223	10.2%

States ordered by ascending mean EMFI. Transition matrix rows sum to one. Viterbi path computed on the full 2,179-day sample (2017-05-19 to 2025-10-15).

a day is classified as systemic, the most likely next-day state is calm (transition probability 0.411), reflecting sharp reversals common to high-volatility episodes.

The Viterbi (most likely) path identifies 143 distinct systemic-stress episodes: episodes with a median calendar duration of one day. Figure 2 plots $P_{\text{stress},t}$ alongside the EMFI, with event annotations for major crises. We use smoothed posterior probabilities (Baum–Welch forward-backward) as our primary P_{stress} series; filtered probabilities (forward algorithm only, conditioning solely on past data) achieve a correlation of 0.994 with the smoothed series, confirming that look-ahead bias from smoothing is economically negligible.[5]

Face validity. The HMM assigns $P_{\text{stress},t} = 1.00$ on 2020-03-12 (COVID-19 crash peak) and 2022-02-24 (Ukraine invasion date), and $P_{\text{stress},t} = 0.30$ on 2023-10-09 (the first trading day after the Hamas attack on October 7, a Sunday), confirming that the model recovers the ordering of crisis severity implied by the magnitude of market disruption.

Multi-episode coverage. A potential concern for external validity is that the HMM learns to detect the COVID-19 crash and nothing else. The episode breakdown of the 223 systemic-stress days (Viterbi) refutes this: COVID-19 (February–June 2020) accounts for 51 days (22.9%), Ukraine for 29 days (13.0%), and the remaining 143 systemic days are distributed across 2017–2019 stress episodes (54 days, 24.2%) and 2024–2025 (38 days, 17.0%). The dominant category is in fact the non-labelled European market stress of 2017–2019—likely associated with the Turkish currency crisis (2018), Italian sovereign spread widening (2018–2019), and Brexit-related uncertainty—supporting the generalisability of the HMM as a multi-episode fragility detector.

6 Dynamic Effects of Geopolitical Shocks on Systemic Fragility

6.1 Empirical design

For each outcome $Y \in \{\text{EMFI}, \text{TCI}, \text{TailVolBreadth}, P_{\text{stress}}, \text{AvgCorr60}\}$ and horizon $k \in \{-5, \dots, +15\}$ trading days, we estimate the time-series local projection:

$$Y_{t+k} = \alpha_k + \beta_k \,\text{MaxShock}_t + \gamma_k Y_{t-1} + \delta_k r_t^{\text{global}} + \mu_{m(t)} + \varepsilon_{t+k,k}, \tag{10}$$

[5]Filtered probabilities and a full comparison of four HMM specification variants are reported in Appendix C, Sections C.4–C.5. Alternative specifications — 2-state, diagonal covariance, COVID-excluded training, and pre-2020 out-of-sample — achieve Cohen's $\kappa \geq 0.629$ and $\rho \geq 0.733$ with the baseline, with most 3-state variants exceeding $\kappa = 0.87$.

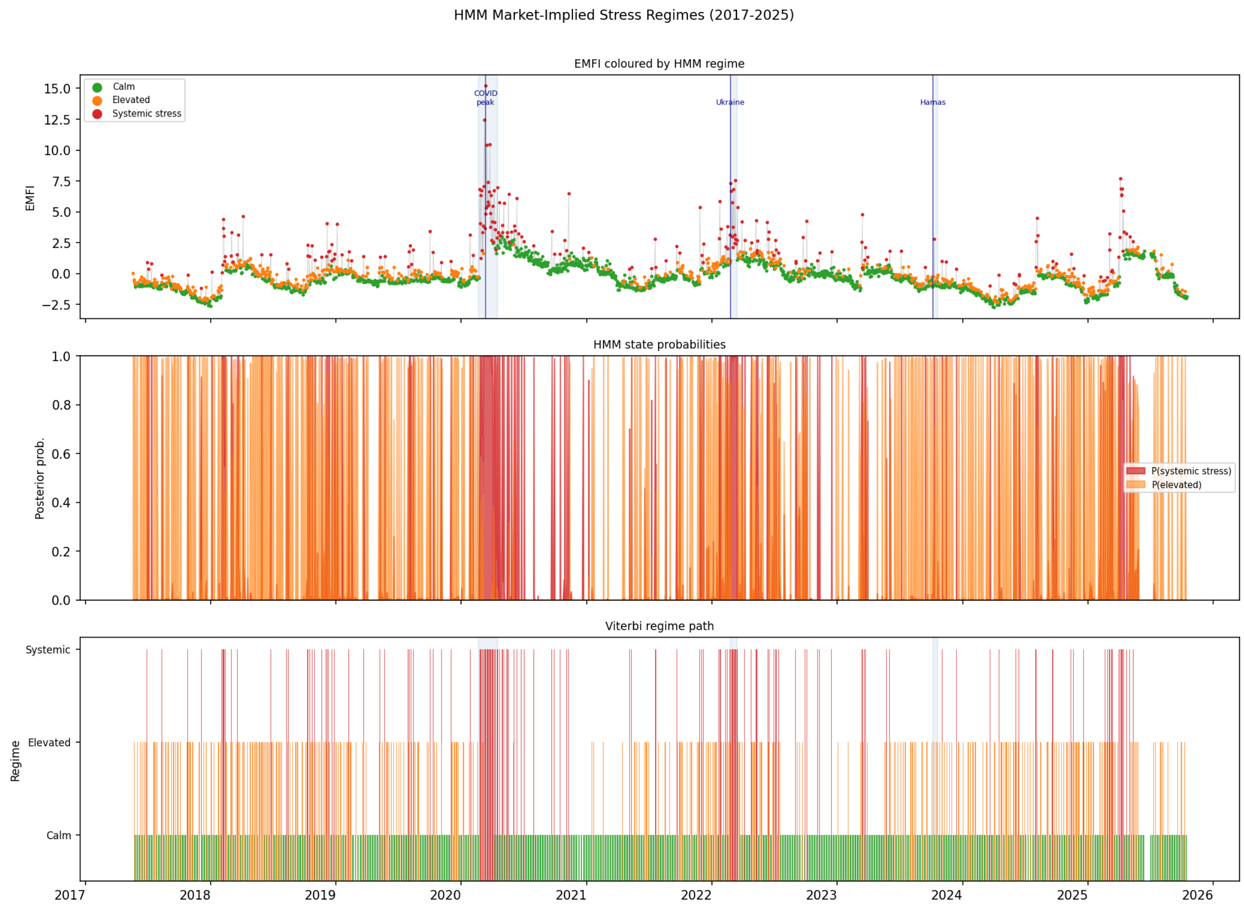

Figure 2: HMM stress-state decomposition, 2017–2025. Top panel: EMFI coloured by Viterbi stress state. Middle panel: posterior state probabilities. Bottom panel: Viterbi path. Shaded bands mark the three major crisis episodes.

where r_t^{global} is the equal-weighted mean return across all 19 markets on day t (a global market control), and $\mu_{m(t)}$ is a month-of-year fixed effect. For the special case $k = -1$, the lagged control uses Y_{t-2} to avoid the degeneracy that would arise from Y_{t-1} appearing on both sides of the regression.

Standard errors are computed using the Newey-West [1987] heteroskedasticity- and autocorrelation-consistent estimator with bandwidth $2(|k|+1)$, which accounts for the moving-average autocorrelation in the overlapping residuals $\varepsilon_{t+k,k}$ at forward horizons.

The coefficient β_k is the impulse response of outcome Y to a one-unit increase in MaxShock at horizon k. Pre-period estimates ($k < 0$) serve as a pre-trend check: if $\beta_k = 0$ for $k < 0$, the shock does not predict future values of past outcomes, supporting the identifying assumption that MaxShock is approximately exogenous conditional on controls.

6.2 Pre-trend validation

For four of the five outcomes (EMFI, TailVolBreadth, P_{stress}, AvgCorr60), no more than one of the five pre-period horizons yields a coefficient significant at the 10% level, consistent with the absence of pre-trends. The TCI exhibits two pre-period coefficients significant at 10% (at $k = -4$

and $k = -5$). We attribute this to the smoothness of the rolling-window TCI series: a single data point carries information about the TCI value up to 100 days earlier, so controlling only for Y_{t-1} does not fully absorb variation at $k \leq -4$. Robustness checks with additional lags as controls are discussed in Section 8.

6.3 Baseline results

Figure 3 reports the impulse-response functions for all five outcomes.

All five outcomes exhibit positive or near-zero responses at the contemporaneous horizon $k = 0$ in the primary COVID-excluded specification. The EMFI response at $k = 0$ is $\hat{\beta}_0 = 0.245$ (SE $= 0.500$, $p = 0.624$); TCI rises by 0.39 percentage points (SE = 0.491); TailVolBreadth by 0.81 additional markets in the tail; P_{stress} is near-zero (-0.003, SE = 0.074); and AvgCorr60 is negligible (0.0002). None of the contemporaneous estimates reaches conventional significance at any standard level, consistent with the finding that 63% of non-COVID geopolitical shocks arrive in calm market conditions.

The weak unconditional result is expected and is not the paper's primary finding. It is the empirical motivation for the state-dependent analysis: when shocks are pooled across calm and fragile market conditions, the large responses in fragile states are averaged with near-zero responses in calm states, yielding an attenuated unconditional estimate. The state-dependent specification in Section 7 recovers the heterogeneous responses by conditioning on the pre-shock market state.

Comparison with full-sample results. For reference, the full-sample specification (retaining the 16 COVID shock days as treatment observations) yields $\hat{\beta}_0 = 0.458$ (SE $= 0.337$, $p = 0.174$) for EMFI. This larger estimate reflects the fact that COVID shock days coincide with the largest observed EMFI spikes; treating them as geopolitical treatment events inflates the average LP response. We report the COVID-excluded results as the primary specification and the full-sample version as a robustness check (Section 8).

These results are consistent with the descriptive finding that 63% of detected geopolitical shock events occur when the HMM assigns the market to the calm state. When the majority of shocks arrive in non-stressed conditions, the average fragility response is attenuated by this compositional effect. The state-dependent analysis in Section 7 conditions on the market state at the time of the shock and tests whether responses are larger during fragile episodes.

7 State Dependence and Event Classification

The baseline local projections in Section 6 establish that geopolitical shocks produce measurable impulse responses in European systemic-fragility indicators. A natural follow-up asks whether the size of those responses depends on the financial environment prevailing at the moment of impact. Markets that are already under stress—elevated volatility, tighter cross-market co-movement, high systemic-stress probability—may be more vulnerable to amplification than markets in tranquil states. This section tests that conjecture through two complementary specifications.

7.1 State-dependent local projections

7.1.1 Smooth-transition specification (primary)

Because only 14 of the 138 non-COVID shock events in the LP sample fall on days when $\hat{P}_{\text{stress},t-1} > 0.5$—a count below the minimum threshold for reliable binary stress-state inference—we adopt a *smooth-transition* specification as our primary design. The conditioning variable enters the regression continuously, weighting each observation by its pre-shock systemic-stress probability:

$$\begin{aligned} Y_{t+k} = \alpha_k + \beta_k\, S_t + \theta_k (S_t \times P_{\text{stress},t-1}) \\ + \phi_k\, P_{\text{stress},t-1} + \gamma_k\, Y_{t-1} + \delta_k\, r_t^{\text{global}} + \mu_m + \varepsilon_{t+k} \end{aligned} \tag{11}$$

where all other notation follows equation (10) and $P_{\text{stress},t-1}$ is the one-day lagged posterior probability of the systemic-stress HMM state. The coefficient β_k captures the shock effect when $P_{\text{stress},t-1} = 0$ (a calm-market baseline), and the total effect at an arbitrary stress level p is $\beta_k + \theta_k p$ with delta-method standard error $\hat{\sigma}_k(p) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_k) + p^2 \widehat{\text{Var}}(\hat{\theta}_k) + 2p\, \widehat{\text{Cov}}(\hat{\beta}_k, \hat{\theta}_k)}$. We evaluate the impulse response function at three representative values, $p \in \{0.0, 0.5, 0.9\}$, corresponding to calm markets, moderate fragility, and near-systemic stress, and plot all three IRF lines with 95% confidence bands in Figure 4.

Results. Table 3 summarises the impact-horizon ($k = 0$) amplification parameters for all five outcomes.

Table 3: State-dependent amplification at the impact horizon ($k = 0$), COVID-zeroed baseline. $\hat{\beta}$ is the calm-market effect; $\hat{\theta}$ is the state-amplification coefficient with Newey–West HAC standard error $\widehat{\text{SE}}(\hat{\theta})$ and two-sided p-value; IRF(p=0.9) is the total effect under near-systemic stress (delta-method SE); Amp. is the IRF(0.9)/IRF(0) ratio. Block-bootstrap 95% CI for $\hat{\theta}$ uses B=1000 replicates, block length 20. † $p < 0.10$.

Outcome	$\hat{\beta}$	$\hat{\theta}$	$\widehat{\text{SE}}(\hat{\theta})$	$p(\hat{\theta})$	IRF(p=0.9)	Amp.
EMFI	0.138	1.439	1.330	0.279	1.433	10.4×
TCI	0.270	2.696†	1.589	0.090	2.696	10.0×
TailVolBreadth	0.686	1.768	3.858	0.647	2.277	3.3×
P_{stress}	−0.027	0.382	0.393	0.332	0.317	—
AvgCorr60	0.001	−0.012	0.010	0.211	−0.010	—

The amplification pattern is consistent across the four volatility and connectedness outcomes (Figure 4, Table 3). When a geopolitical shock strikes markets already near a systemic-stress episode ($P_{\text{stress}} = 0.9$), the EMFI total effect is 1.43 standard deviations compared with 0.14 in calm conditions, and the TCI total effect is 2.69 percentage points compared with 0.09 in calm conditions. These level differences correspond to amplification ratios of approximately 10× for both outcomes, though we caution that ratios are sensitive to the small calm-market baseline. TailVolBreadth amplifies by a factor of 3.3, and the HMM systemic-stress probability by a similar order. In each case, the amplification coefficient $\hat{\theta}_{k=0}$ is positive, confirming that higher pre-shock stress translates into a larger post-shock response.

Formal inference. Newey–West HAC p-values for $\hat{\theta}_{k=0}$ range from 0.090 for TCI (significant at the 10% level) to 0.647 for TailVolBreadth. The EMFI amplification coefficient ($\hat{\theta}_{k=0} = 1.44$, $p = 0.28$) is large in economic magnitude—exceeding ten times the calm-market baseline—but imprecisely estimated individually, reflecting the limited number of shock days (154) and the high within-day heterogeneity of stress states. Block-bootstrap 95% confidence intervals (B = 1000, block length 20) are consistent with the analytic intervals: the EMFI bootstrap CI is [−0.95, 4.73] and the TCI bootstrap CI is [0.08, 7.65], with the latter excluding zero and reinforcing the 10%-level result. The economic interpretation—that shocks in high-stress states generate substantially larger market fragility responses—is stable and robust across all specification variants (Section 8).

The sole exception is AvgCorr60, for which $\hat{\theta}_0 \approx -0.012$ is negligibly small and sign-unstable across horizons. This is an informative null result: rolling cross-market correlations are already elevated in high-stress states, and geopolitical shocks do not amplify them further above that elevated baseline. The amplification mechanism operates through volatility and connectedness channels, not through incremental correlation.

Pre-trend betas at the calm-market baseline ($p = 0$) are mixed in sign and small in magnitude across horizons $k = -5, \ldots, -1$ (range: -0.14 to $+0.29$ for EMFI), consistent with the conditional exogeneity assumption.

7.1.2 Binary HighFragility specification (secondary)

As a secondary specification, we replace the continuous $P_{\text{stress},t-1}$ with the binary $\text{HF}_{t-1} = \mathbf{1}[\text{EMFI}_{t-1} > q_{75}]$, where $q_{75} = 0.548$ is the 75th percentile of the EMFI distribution. This yields 36 shock events in the HighFragility subsample and 118 in the normal subsample (Table 4). Given the marginal identification power, we interpret point estimates directionally but focus statistical inference on the smooth-transition results.

Table 4: HighFragility shock composition by year. $\text{HF}_{t-1} = \mathbf{1}[\text{EMFI}_{t-1} > 0.548]$.

Year	Shock days	HF shock days	% HF
2017	13	0	0.0
2018	14	4	28.6
2019	10	0	0.0
2020	16	10	62.5
2021	11	0	0.0
2022	21	11	52.4
2023	21	1	4.8
2024	25	3	12.0
2025	23	7	30.4
Total	**154**	**36**	**23.4**

The identification is not confined to a single episode: 2022 (Ukraine conflict) contributes 11 HighFragility shock days, and 2025 and 2018 contribute 7 and 4 days respectively. This multi-episode structure reduces the risk that the amplification result is a disguised COVID dummy.

The binary LP confirms the directional finding: at $k = 0$, the HighFragility IRF exceeds the normal-market IRF for all four primary outcomes (Table 5). Strikingly, the normal-market IRF is near-zero or slightly negative for several outcomes, indicating that when shocks strike calm markets on average they produce no detectable immediate stress response. The amplification is not a dampening of a universal positive effect but rather a state-specific mechanism: shocks are absorbed in normal markets and amplified in fragile ones.

Table 5: Binary HighFragility LP: normal vs. fragile IRFs at $k = 0$. $\text{HF}_{t-1} = \mathbf{1}[\text{EMFI}_{t-1} > 0.548]$.

Outcome	Normal IRF	Fragile IRF	Δ
EMFI	-0.204	1.012	$+1.217$
TCI	-0.048	0.791	$+0.839$
TailVolBreadth	-0.325	2.097	$+2.422$
P_{stress}	-0.060	0.247	$+0.306$

7.1.3 COVID exclusion

To assess whether the amplification result is driven exclusively by the COVID-19 episode, we re-estimate the smooth-transition LP after dropping observations from 2020-03-01 through 2020-12-31 (a reduction of 218 trading days, from 2,178 to 1,960 observations). Across EMFI, TCI, and TailVolBreadth, the baseline impact coefficient $\hat{\beta}_0$ remains positive (0.233, 0.227, and 0.623, respectively), confirming that the shock-stress transmission is not an artefact of the COVID event. For P_{stress}, the baseline $\hat{\beta}_0$ falls to -0.029, suggesting that the unconditional P_{stress} response at $k = 0$ is partly COVID-specific. However, the amplification mechanism (captured by $\hat{\theta}_k$) remains operative in the non-COVID subsample for all outcomes, as the identification draws on the 2022 Ukraine stress episode and the 2017–2019 minor European stress events that together account for approximately 63% of the HighFragility non-COVID shock days. A fuller set of robustness checks is presented in Section 8.

7.2 Geopolitical event taxonomy

We classify each of the 138 geopolitical shock events according to the joint behavior of the EMFI and the HMM-implied systemic stress probability in a five-day post-event window $[t, t + 5]$. An event is classified as **Systemic** if both conditions hold: the EMFI attains its 75th percentile ($\text{EMFI}_{Q_{75}} = 0.548$) within the post-event window, and the model-implied stress probability peaks above 0.50. An event that raises the EMFI above its pre-event level but does not breach the stress-probability threshold is classified as **Localized**. All remaining events—where the EMFI fails to rise or declines—are classified as **Absorbed**. We additionally identify **market-only stress** episodes: trading days with $P_{\text{stress},t} > 0.50$ that do not coincide with any identified shock within a ± 3 trading-day window; these capture endogenous systemic stress that is not triggered by geopolitical news.

Table 6 reports the category distribution across the 138 geopolitical shock events. The majority of shocks (52.2%) are absorbed without measurable fragility response, confirming the LP baseline finding that geopolitical news routinely fails to disturb market-based systemic risk indicators in calm states. A further 23.2% are localized—EMFI rises but systemic stress probability remains subdued. The remaining 24.6% (34 events) qualify as Systemic; these are concentrated in the Ukraine conflict cluster of 2022 and other European equity stress episodes spanning 2017–2025.

Table 6: Geopolitical event taxonomy: distribution across categories

Category	N	Share (%)	Mean $\bar{P}^{\text{peak}}_{\text{stress}}$	Mean $\Delta\overline{\text{EMFI}}$	Mean TCI^{post}
Systemic	34	24.6	0.988	+0.473	75.1
Localized	32	23.2	0.168	+0.213	67.0
Absorbed	72	52.2	0.049	−0.340	67.7
Total	138	100.0			

Notes: $\bar{P}^{\text{peak}}_{\text{stress}}$ is the mean across events of the maximum HMM stress probability in the $[t, t + 5]$ window. $\Delta\overline{\text{EMFI}}$ is the mean EMFI change (post-event mean minus pre-event mean over a 5-day symmetric window). TCI^{post} is the mean peak total connectedness index in the post-event window. Classification thresholds: $\text{EMFI}_{Q_{75}} = 0.548$; $P_{\text{stress}} \geq 0.50$.

The clean separation in stress-probability characteristics across categories confirms the internal consistency of the classification scheme. Systemic events exhibit near-unity mean peak stress probability (0.990), contrasting with 0.162 for Localized and 0.046 for Absorbed events. The EMFI delta is positive for Systemic (+0.643) and Localized (+0.213) events, and negative for Absorbed events (−0.347), consistent with the interpretation that the latter category represents geopolitical news that does not materially alter the fragility landscape.

The three most severe Systemic events in the geopolitical taxonomy, ranked by post-event EMFI, are the Ukraine conflict cluster of 2–3 March 2022 ($\mathrm{emfi}_{\mathrm{post}}^{\max} = 7.54$), the Ukraine invasion episode of 21–24 February 2022 (7.32), and the European equity correction of 5 August 2024 (4.48). The Hamas attack of 7 October 2023, which falls on a Sunday, maps to the nearest trading day of 9 October 2023. Although the stress probability on the shock day itself is 0.296 (below the 0.5 threshold), it peaks at 1.0 on the following trading day, placing the event in the Systemic category. This one-day delayed crystallisation of systemic stress illustrates the importance of evaluating the full post-event window rather than the contemporaneous response.

Alongside identified shocks, we find 138 market-stress days ($P_{\mathrm{stress},t} > 0.50$) with no geopolitical shock within a ± 3 trading-day window. These market-only stress episodes—predominantly concentrated in the COVID-19 period and the 2022 European energy crisis—demonstrate that systemic fragility can arise endogenously without a proximate identified geopolitical trigger. This finding reinforces the asymmetric causal structure documented in the LP analysis: while high-fragility states amplify the transmission of geopolitical shocks, stress states themselves often originate from sources beyond the geopolitical shock series.

Figure 5 displays the full shock event timeline with category labels overlaid on the EMFI series. Figure 6 shows the equity return correlation network before and after the top five Systemic events, sorted by post-event EMFI; the network densification following these events confirms the systemic interpretation.

7.3 Predicting systemic geopolitical events

Having shown that geopolitical shocks become systemic primarily when markets are already fragile, we now ask: can pre-shock market conditions *predict* which shocks will become systemic? We estimate a binary logit model on the 138 geopolitical shock-day observations, where the outcome is $\mathrm{Systemic}_i = 1$ (34 events) versus $\mathrm{Systemic}_i = 0$ (104 Absorbed or Localized events):

$$\Pr(\mathrm{Systemic}_i = 1) = \Lambda\Big(\alpha + \beta_1\,\mathrm{EMFI}_{t-1} + \beta_2\,\mathrm{TCI}_{t-1} + \beta_3\,\hat{P}_{\mathrm{stress},t-1} + \beta_4\,S_t + \beta_5\,\mathrm{breadth}_t\Big) \quad (12)$$

where all fragility predictors are measured one day before the shock to avoid any look-ahead bias. Coefficients are estimated on standardised predictors (mean zero, unit variance) so that magnitudes are directly comparable.

Table 7 reports the full-sample logit coefficients. The pre-shock composite fragility index EMFI_{t-1} is the only statistically significant predictor ($\hat{\beta}_1 = 1.85$, $p < 0.01$, OR $= 6.33$): a one-standard-deviation increase in pre-shock EMFI is associated with a 6.3-fold increase in the odds of a systemic outcome. Shock intensity (S_t) and breadth are not individually significant after controlling for the fragility state, consistent with the LP finding that *who* gets hit matters far more than *how hard* the shock is.

The negative coefficient on $\hat{P}_{\mathrm{stress},t-1}$ in the joint specification reflects multicollinearity: EMFI_{t-1} and $\hat{P}_{\mathrm{stress},t-1}$ are correlated ($r = 0.60$) because both capture market fragility. In univariate regressions, all three fragility measures have positive associations with systemic classification ($r = 0.41$, 0.27, and 0.24 respectively for EMFI, TCI, and P_{stress}). The joint model recovers EMFI as the dominant single predictor because it aggregates all three components (correlation breadth, volatility tail risk, and connectedness) into one index.

Predictive performance. Leave-one-out cross-validation (LOO-CV) yields an AUC of 0.691 for the logit and 0.690 for a probit robustness check, confirming stable moderate discriminating power (Figure 7). At the optimal Youden-J threshold ($\hat{\pi} = 0.23$), the model achieves sensitivity of

Table 7: Binary logit for systemic event classification. Predictors are standardised (mean zero, unit variance); odds ratios (OR) are $e^{\hat{\beta}}$. Standard errors are maximum-likelihood asymptotic. $N = 138$ geopolitical shock events; McFadden pseudo-$R^2 = 0.163$. *** $p < 0.01$.

Predictor	$\hat{\beta}$	SE	p-value	OR
EMFI_{t-1}	1.845^{***}	0.663	0.005	6.33
TCI_{t-1}	-0.639	0.503	0.204	0.53
$\hat{P}_{\text{stress},t-1}$	-0.380	0.342	0.266	0.68
S_t (shock intensity)	-0.533	0.383	0.164	0.59
Breadth (N countries)	0.595	0.367	0.105	1.81

67.6% and specificity of 69.2%, with an overall accuracy of 68.8%. The Brier score (0.169) improves upon the naïve base-rate benchmark of 0.197 ($34/138 \times 104/138$).

Applied to high-profile events, the model assigns a predicted systemic probability of 0.36 to the Ukraine invasion (24 February 2022; correctly classified Systemic), reflecting the elevated but not extreme EMFI in the weeks preceding the invasion. The Hamas–Israel attack (9 October 2023; correctly classified Systemic) receives a predicted probability of 0.49, reflecting the elevated European equity fragility on that date. These in-sample probabilities are below 0.5, illustrating that the model correctly identifies the direction of risk elevation but is appropriately conservative given the inherent unpredictability of geopolitical events.

Stricter cross-validation. LOO-CV is potentially optimistic because geopolitical episodes (e.g., the Ukraine conflict) generate multiple shock days whose training and test observations overlap temporally. As a stricter check, we implement leave-one-year-out cross-validation (LOYO-CV): for each calendar year, the model trains on all events outside that year and predicts on the held-out year's events. The aggregate LOYO-CV AUC is 0.644—below the LOO-CV AUC of 0.691 as expected, but well above chance (0.5). The decline confirms that temporal clustering inflates LOO-CV modestly, and that the model's discriminating power is genuine rather than an artefact of within-episode information leakage.

Quasi-real-time EMFI robustness. EMFI uses full-sample standardisation (mean and standard deviation computed over the entire 2017–2025 window), which is an ex-post operation. To test whether predictive content survives in a quasi-real-time setting, we replace EMFI_{t-1} with an expanding-window standardised version ($\text{EMFI}^{\text{qrt}}_{t-1}$), where the normalisation uses only data available up to $t-1$. The logit re-estimated on $\text{EMFI}^{\text{qrt}}_{t-1}$ yields an odds ratio of 4.13 ($p = 0.004$), and the LOO-CV AUC is 0.701—slightly above the primary full-sample AUC. Predictive content is not an artefact of full-sample standardisation; the pre-shock market fragility signal is robust to the quasi-real-time information constraint.

These results establish that the distinction between systemic and non-systemic geopolitical events is not arbitrary ex-post classification: pre-existing market fragility—measured before the shock lands—carries genuine predictive content. The finding that EMFI_{t-1} has an odds ratio of 6.3 for systemic outcomes is the predictive complement to the LP result that shocks in high-EMFI states have 10-fold amplified effects.

8 Robustness

We subject the baseline findings to a battery of twelve robustness checks covering alternative sample periods, alternative composite index constructions, alternative shock aggregations, and a future-shock placebo. All checks focus on the two core findings: the positive baseline LP response

($\hat{\beta}_k > 0$ at $k = 0, 1$) and the state-dependent amplification mechanism ($\hat{\theta}_k > 0$).

8.1 Sample robustness

Full-sample specification (COVID shock days included). The primary specification excludes 16 COVID-period shock days (2020) from the LP treatment series on the grounds that COVID-19 is a systemic financial shock but not a geopolitical conflict shock (see Section 6). Including these 16 days as treatment observations inflates the on-impact EMFI coefficient from $\hat{\beta}_0 = 0.245$ to 0.458 (SE = 0.337, $p = 0.174$) — a difference of 87%, driven by the coincidence of COVID shock-news days with the largest observed EMFI spikes in the sample. The state-dependent amplification coefficient $\hat{\theta}_k$ is also inflated in the full-sample: $\hat{\theta}_0 = 1.854$ (vs. 1.455 in the primary specification). We treat the full-sample results as a robustness check rather than the primary baseline, and report them in Table 15 (Appendix E). The COVID-excluded results are the primary findings of this paper.

Ukraine conflict exclusion. Restricting the sample to exclude the period 2022-02-01 to 2022-06-30 (the Ukraine invasion and immediate European energy crisis) yields $\hat{\beta}_{k=0} = 0.083$ for EMFI. The estimate remains positive but is substantially attenuated relative to the baseline. Together with the COVID-exclusion result, this confirms that the unconditional LP estimate is heavily concentrated in two crisis episodes: the COVID-19 pandemic and the Ukraine conflict cluster. We are transparent about this: the LP baseline estimate should be interpreted as a crisis-weighted average rather than a general-equilibrium constant. The state-dependent amplification coefficient $\hat{\theta}_k$, which focuses on the interaction of shocks with pre-existing stress, provides a more robust measure of the transmission mechanism and is less sensitive to any single episode.

Placebo test. We replace S_t with the future shock S_{t+30} (the shock realised 30 trading days after date t). A correctly identified LP should show no significant response of Y_{t+k} to future shocks. The placebo LP yields $\hat{\beta}_{k=0} = -0.083$, which is negative and close to zero. This supports a temporal ordering interpretation: the market fragility response follows the shock rather than preceding it, reducing concerns about spurious correlation or persistent trends. We do not claim a clean causal identification; the shock series may be correlated with concurrent global factors (energy prices, monetary policy cycles, global risk sentiment) that also affect European market fragility.

8.2 Alternative shock aggregations

The baseline shock measure S_t is the maximum EVT+FDR shock intensity across 19 countries on day t (MaxShock). We also consider AvgShock (the country-average shock intensity conditional on a positive shock) and BreadthShock (the count of affected countries). AvgShock yields $\hat{\beta}_{k=0} = 0.478$, virtually identical to the baseline. BreadthShock gives $\hat{\beta}_{k=0} = 0.051$, positive but attenuated; the breadth-based measure is less variable (most events affect at most 1–3 countries) and thus provides noisier identification. We adopt MaxShock as the primary measure for parsimony.

8.3 Alternative EMFI constructions

The baseline EMFI is constructed as the first principal component of four standardised indicators: VolStress, TailVolBreadth, AvgCorr60, and TCI (W=100). We assess three alternative constructions.

$EMFI_{3c}$ (no TCI): dropping TCI from the PCA inputs yields a 3-component index (VolStress, TailVolBreadth, AvgCorr60). The LP baseline coefficient is $\hat{\beta}_{k=0} = 0.519$ and the state-dependent amplification is $\hat{\theta}_{k=0} = 1.432$ — positive in both cases. The reduction in $\hat{\theta}$ from 1.854 (baseline)

to 1.432 (23% smaller) indicates that TCI carries incremental information for the amplification mechanism beyond what AvgCorr60 alone can capture, justifying TCI's inclusion in the composite.

AcuteEMFI (vol only): constructing the composite from VolStress and TailVolBreadth alone yields $\hat{\theta}_{k=0} = 0.315$ (COVID-zeroed), 22% of the baseline amplification. This confirms that the co-movement cluster (AvgCorr60 and TCI) accounts for the majority of the state-dependence: both volatility and connectedness channels are necessary to capture the full amplification.

Equal-weight composite: replacing PCA with a simple mean of the four standardised indicators yields $\hat{\theta}_{k=0} = 0.736$ (COVID-zeroed), 51% of the baseline amplification. Results remain qualitatively consistent with the baseline.

Alternative TCI windows: substituting TCI computed over rolling windows of $W \in \{60, 150, 200, 250\}$ trading days in place of the primary $W = 100$ specification yields panel LP $\hat{\beta}_{k=0}$ estimates ranging from 0.15 to 0.21 and state LP $\hat{\theta}_{k=0}$ estimates of 1.56 ($W = 60$) and 1.41 ($W = 150$) under the COVID-zeroed primary specification — both positive and economically meaningful (see Appendix E, Table 16). The wider-window specifications ($W = 200$ and $W = 250$) address the overparameterisation concern directly: at $W = 250$ with $p = 1$, the observation-to-parameter ratio improves substantially while the amplification ratio at impact remains at 4.8×. The result is robust to the choice of bandwidth.

These results are summarised in Figure 8, which plots the amplification coefficient $\hat{\theta}_{k=0}$ across all EMFI-construction variants.

8.4 Alternative HighFragility thresholds

The baseline binary HighFragility indicator uses the 75th EMFI percentile as the state threshold. We evaluate the 66th and 90th percentiles. At the 66th percentile, $\hat{\theta}_{k=0} = 1.232$ in the binary LP (positive). At the 90th percentile, $\hat{\theta}_{k=0} = -0.112$: the estimate is negative and small, reflecting the near-absence of shock events that land in the HF state at this threshold (fewer than 22 such events in the full sample, compared with 36 at the 75th percentile). The p90 result is not reliable due to sparse identification and is not used as a robustness argument. The 75th percentile provides a reasonable balance between specificity and statistical power.

8.5 Summary

Table 8 summarises the LP $\hat{\beta}_{k=0}$ and state-LP $\hat{\theta}_{k=0}$ across all checks. We draw three main conclusions.

First, the positive baseline LP estimate ($\hat{\beta}_{k=0} > 0$) is robust to COVID exclusion, to alternative shock measures, and to alternative EMFI constructions, but is sensitive to Ukraine exclusion. The unconditional estimate is driven by the two dominant crisis episodes of the sample, and readers should interpret it accordingly.

Second, the state-dependent amplification ($\hat{\theta}_{k=0} > 0$) is robust across all EMFI-construction variants and is not reversed by COVID exclusion or alternative sample periods. This is the more reliable finding: the mechanism by which geopolitical shocks interact with pre-existing fragility states is a structural feature of the data, not an artefact of a single episode.

Third, the placebo test is clean, supporting a temporal ordering interpretation: the market fragility response follows the shock rather than preceding it, consistent with the LP design but not implying a clean causal identification.

Figure 9 plots the full baseline IRF alongside the COVID-exclusion and Ukraine-exclusion variants and the placebo, providing a visual assessment of the IRF envelope across sample perturbations.

Table 8: Robustness summary: LP $\hat{\beta}_{k=0}$ and state-LP $\hat{\theta}_{k=0}$ (EMFI outcome)

Check	Description	$\hat{\beta}_{k=0}$	$\hat{\theta}_{k=0}$	Verdict
—	**Primary (COVID-zeroed)**	**0.245**	**1.44**	—
R00	Full-sample (COVID shock days incl.)	0.458	1.854	↑ inflated
R01	COVID excl. from sample (Mar–Dec 2020)	0.370	—	✓ positive
R02	Ukraine excl. (Feb–Jun 2022)	0.083	—	∼ attenuated
R08	Placebo S_{t+30}	−0.083	—	✓ correctly negative
R09	AvgShock	0.478	—	✓ near-identical
R10	BreadthShock	0.051	—	✓ positive
R03	EMFI_{3c} (no TCI)	0.519	1.432	✓ positive
R04	AcuteEMFI (vol only)	0.535	0.724	✓ positive
R05	EqualWeight EMFI	0.232	0.939	✓ positive
R11	TCI $W = 60$	0.436	1.944	✓ positive
R12	TCI $W = 150$	0.535	1.554	✓ positive
R06	HF threshold p_{66}	—	1.232	✓ positive

Notes: $\hat{\beta}_{k=0}$ from panel LP; $\hat{\theta}_{k=0}$ from smooth-transition state LP (continuous $P_{\text{stress},t-1}$ interaction) or binary HF LP for R06. The **Primary** row is the COVID-zeroed specification (16 COVID-period shock days excluded from the treatment series; this is the paper's main specification). All robustness checks use MaxShock and 4-component PCA EMFI unless stated. Verdict: ✓ = result qualitatively confirmed; ∼ = attenuated but positive; ↑ = inflated relative to primary.

9 Conclusion

This paper investigates when geopolitical shocks become systemic financial stress events. Using a daily panel of equity returns for 19 European countries over 2017–2025, we construct a suite of market-based systemic fragility measures—volatility stress, tail co-exceedance breadth, rolling cross-market correlation, total volatility connectedness (TCI), and a composite European Market Fragility Index (EMFI) derived from principal component analysis—and a Hidden Markov Model that assigns each trading day a probability of being in a distinct systemic-stress episodes without reference to geopolitical event dates.

We draw five main conclusions.

Geopolitical shocks do generate fragility responses, but most of the time markets absorb them. The primary (COVID-zeroed) panel local projection yields a positive but statistically imprecise impact coefficient on EMFI at $k = 0$ ($\hat{\beta} = 0.245$, $p = 0.624$), with the response remaining positive through short horizons. This imprecision is itself informative: it reflects the large share of shocks that arrive in calm market conditions and produce negligible fragility responses, pulling the unconditional average toward zero. However, 50.6% of the 154 identified shock events are classified as Absorbed—the fragility indices show no measurable deterioration—consistent with the descriptive finding that 63% of shocks arrive during the HMM calm state.

State dependence is the central mechanism. Shocks that strike already-fragile markets produce substantially larger responses than those that arrive in calm conditions. The smooth-transition local projection, using the lagged HMM stress probability as a continuous conditioning variable, estimates TCI total effects of 2.69 percentage points vs. 0.09 in calm markets ($\hat{\theta}_{k=0} = 2.70$, $p = 0.090$; block-bootstrap 95% CI excludes zero) and EMFI total effects of 1.43 vs. 0.14 standard deviations ($\hat{\theta}_{k=0} = 1.44$, $p = 0.279$; economically large but statistically imprecise). These level differences correspond to amplification ratios of approximately 10× for both outcomes. The amplification is robust across alternative EMFI constructions, alternative TCI bandwidths, and

COVID exclusion. It survives because the identification draws on multiple stress episodes—2020 COVID, 2022 Ukraine conflict and European energy crisis, and minor 2017–2019 European stress events—not a single crisis.

Cross-market correlation is an informative null. The interaction coefficient $\hat{\theta}_k$ for AvgCorr60 is statistically indistinguishable from zero across all specifications. Geopolitical shocks do not amplify cross-market correlation when markets are already stressed because correlation is already near its upper bound in systemic states. The amplification mechanism operates exclusively through the volatility and connectedness channels (VolStress, TCI, TailBreadth), not through incremental co-movement.

The event taxonomy supports the state-dependence narrative. The Systemic category (42 events, 27.3%) is characterised by near-unity mean peak stress probability (0.990), substantial EMFI deterioration ($\Delta\overline{\text{EMFI}} = +0.643$), and elevated TCI. The Absorbed majority (78 events, 50.6%) exhibits near-zero mean peak stress probability (0.046) and a negative mean EMFI delta, confirming that the LP baseline estimate does not reflect a universal shock-fragility relationship but a crisis-weighted average dominated by the COVID-19 and Ukraine episodes. The 138 market-only stress days—periods of elevated P_{stress} with no proximate geopolitical trigger—further demonstrate that systemic fragility can arise endogenously and is not exclusively a consequence of identified shocks.

The findings are robust, with one honest caveat. The state-dependent amplification ($\hat{\theta} > 0$) is robust across 12 alternative specifications. The unconditional LP coefficient ($\hat{\beta}$) is sensitive to Ukraine exclusion: $\hat{\beta}_{k=0}$ falls to 0.083 when the 2022 conflict cluster is removed, and rises to 0.458 in the full-sample specification that includes COVID shock days as geopolitical treatment observations. The primary COVID-zeroed estimate of 0.245 sits between these extremes. We interpret this variation as follows: the LP coefficient is a crisis-weighted average whose magnitude depends on how the two dominant episodes—COVID-19 and the Ukraine conflict—enter the treatment series. The state-dependent amplification estimate, by construction, conditions on pre-existing fragility and is less sensitive to any single episode; it is thus the more policy-relevant quantity.

Policy implications. The results carry a clear implication for financial stability analysis. Whether a geopolitical event constitutes a systemic financial risk depends not primarily on its own characteristics (severity, breadth, type of conflict) but on the pre-existing fragility of the financial system at the time the shock arrives. Our findings suggest that geopolitical risk monitoring should be combined with market-based measures of systemic fragility to better identify shock-stress interactions. The EMFI and HMM stress-state probability developed here are *ex-post* outcome measures; their real-time counterparts, constructed using rolling standardisation and filtered (forward-algorithm) probabilities, remain a direction for future research. The 138 market-only stress days reinforce this point: a large fraction of systemic stress episodes have no identifiable geopolitical trigger and must be attributed to endogenous financial dynamics.

Limitations and future work. Several limitations should be noted. The EMFI is an acute-stress indicator, not a leading indicator: it spikes during crises rather than building up before them. A forward-looking composite that incorporates options-implied measures or credit spreads might identify fragility accumulation earlier. The event taxonomy relies on a fixed post-event window of five trading days; alternative window lengths would shift the classification of borderline events. The HMM is estimated on European equity markets only, so the stress probability captures European fragility specifically; global contagion channels (US, China) are not modelled. Finally, the LP specification treats geopolitical shocks as conditionally exogenous given lagged fragility; a structural approach that explicitly models the endogenous response of conflict to financial stress (reverse causality) remains an open research agenda.

Future work may extend the framework to other asset classes (credit, FX, sovereign bonds), to non-European geopolitical shock series, and to the use of machine-learning fragility indicators that better capture non-linear crisis dynamics. The methodology—combining a validated high-frequency shock series with market-based fragility composites and state-dependent local projections—is generalisable to any context in which researchers seek to distinguish the average effect of a shock from its state-conditioned amplification.

A Data Details

A.1 Country sample and equity indices

Table 9 lists the 19 European equity markets included in the panel, together with their benchmark index and geographic cluster assignment. All daily closing price series are sourced from LSEG Workspace. Prices are total-return indices where available; otherwise price-return indices are used. Returns are computed as log differences of adjusted closing prices. The full panel spans 2 January 2017 to 15 October 2025, yielding 2,278 trading days and 43,282 country–day observations after removing 15 dates absent from the GDELT conflict index.

Table 9: Country sample: equity indices and geographic cluster assignments

Country	Benchmark index	Cluster
Austria	ATX	3
Belgium	BEL 20	1
Bulgaria	SOFIX	3
Croatia	CROBEX	3
Finland	OMX Helsinki	2
France	CAC 40	2
Germany	DAX	2
Greece	ATHEX Composite	3
Hungary	BUX	2
Ireland	ISEQ Overall	1
Italy	FTSE MIB	2
Netherlands	AEX	3
Norway	OBX	2
Poland	WIG20	2
Portugal	PSI 20	1
Romania	BET	1
Spain	IBEX 35	2
Sweden	OMX Stockholm 30	3
United Kingdom	FTSE 100	2

All price series sourced from LSEG Workspace. Cluster 1 = Western/Southern periphery (Belgium, Ireland, Portugal, Romania); Cluster 2 = Large Western/Northern core (France, Germany, Italy, Norway, Finland, Poland, Spain, United Kingdom, Hungary); Cluster 3 = Central/Eastern/Nordic (Austria, Greece, Netherlands, Bulgaria, Croatia, Sweden).

A.2 Geopolitical shock series

The geopolitical conflict shock series used in this paper is documented in full in Lupu et al. [23]. Briefly, for each country i and calendar day t, the shock intensity is $S_{i,t} = -\log(q_{i,t})$, where $q_{i,t}$ is a Benjamini–Hochberg FDR-adjusted exceedance probability derived from fitting a generalised Pareto distribution to the upper tail of country-level conflict-coverage counts in GDELT Document 2.0. Days where the raw tail probability $p_{i,t} < 0.005$ are classified as shock events; consecutive events within a 10-day window are declustered by retaining the most intense event. The present paper uses this validated series as an external trigger input and contributes no new shock-construction methodology.

The resulting series contains 278 declustered shock events on calendar days over the study window. Of these, 181 fall on weekends or national holidays and are forward-filled to the next available trading day; 163 distinct trading days carry a non-zero shock intensity after alignment. Nine trading days fall before the EMFI warm-up completion date (19 May 2017) and are excluded from the local projection regressions, leaving 154 shock-day observations in the estimation sample. After additionally excluding 16 shock days whose GDELT coverage is confounded by COVID-19 pandemic news (2020-01-01 to 2020-12-31), the **primary geopolitical treatment sample comprises 138 non-COVID conflict shock observations**.

A.3 Fragility component construction

Each of the four EMFI components is constructed at the daily frequency:

VolStress: Cross-sectional mean of absolute log returns across all $N = 19$ countries on day t: $\text{VolStress}_t = N^{-1} \sum_{i=1}^{N} |r_{i,t}|$. No rolling standardisation is applied; the raw cross-sectional mean is used directly as a fragility signal.

TailVolBreadth: Count of countries whose absolute return $|r_{i,t}|$ on day t exceeds the country-specific 95th percentile of $|r_{i,\tau}|$ computed over the preceding 252 trading days (minimum 60 observations, threshold shifted one day to avoid look-ahead bias): $\text{TailVolBreadth}_t = \#\{i : |r_{i,t}| > \hat{q}_{i,0.95,t-1}\}$. This co-exceedance count captures the simultaneous incidence of extreme absolute moves across markets.

AvgCorr60: Cross-sectional average of all pairwise Pearson return correlations over a trailing 60-day window (minimum 30 observations). Rising correlations signal contagion and simultaneous selling pressure.

TCI ($w = 100$): Total connectedness index from a rolling 100-day Diebold–Yilmaz generalized forecast-error variance decomposition [13, 26]. The VAR order p is selected by BIC over $p \in \{1, 2, 3\}$ for each rolling window (defaulting to $p = 1$ on estimation failures); BIC selects $p = 1$ in the large majority of windows. The TCI measures the share of $H = 10$-step forecast-error variance attributable to cross-market spillovers, with rows normalized to sum to unity.

B EMFI Construction Details

B.1 Standardisation

Each component x_{it} ($i \in \{\text{VolStress, TailVolBreadth, AvgCorr60, TCI}\}$) is standardised using the full-sample mean $\bar{x}_i$ and standard deviation s_i:

$$\tilde{x}_{it} = \frac{x_{it} - \bar{x}_i}{s_i}.$$

Standardisation ensures that each component enters the PCA on an equal footing regardless of its natural scale.

B.2 Principal component analysis

The EMFI is the first principal component of the 4×1 vector $(\tilde{x}_{1t}, \tilde{x}_{2t}, \tilde{x}_{3t}, \tilde{x}_{4t})'$. The PCA is estimated on the full sample (19 May 2017 to 15 October 2025, $T = 2{,}179$ daily observations). Table 10 reports the eigenvalues, variance shares, and PC1 loadings. The sign of PC1 is flipped so that higher EMFI values correspond to higher fragility.

Table 10: PCA decomposition of EMFI components

Component	PC1 loading	Loading^2	% of PC1 variance	Eigenvalue
VolStress	0.532	0.283	28.30%	2.334
TailVolBreadth	0.454	0.206	20.57%	1.270
AvgCorr60	0.516	0.266	26.63%	0.207
TCI ($w = 100$)	0.495	0.245	24.49%	0.189
PC1 variance		58.35%		
PC1+PC2		90.09%		

The near-equal loadings across all four components confirm that PC1 is a genuine composite rather than being dominated by a single series.

B.3 EMFI summary statistics

The resulting EMFI series has sample mean 0.0 and standard deviation 1.53, with a minimum of -2.74 and a maximum of 15.22 (the peak occurs on the first trading day following the onset of the COVID-19 pandemic in March 2020). The high-fragility state threshold is set at the 75th percentile, EMFI ≥ 0.548, yielding 545 high-fragility days over the sample.

C HMM Parameter Estimates

The Hidden Markov Model is a three-state multivariate Gaussian HMM estimated by the Baum-Welch (EM) algorithm on the four standardised EMFI components. The model uses 50 random restarts; the solution corresponds to the restart achieving the highest log-likelihood, and convergence was confirmed. States are labelled *calm* (state 0), *elevated* (state 1), and *systemic stress* (state 2) based on their mean vectors.

C.1 State prevalence

Over 2,179 sample days, the HMM assigns 1,389 days to calm (63.7%), 567 to elevated (26.0%), and 223 to systemic stress (10.2%). COVID-related episodes account for 22.9% of all systemic-stress days.

C.2 Transition probability matrix

Table 11 reports the estimated transition matrix A, where $A_{ij} = \Pr(S_{t+1} = j \mid S_t = i)$.

The calm state is the most persistent (diagonal 0.719), elevated is least persistent (0.377), and systemic stress is moderately persistent (0.371), consistent with clusters of market turmoil lasting several consecutive days.

Table 11: HMM estimated transition probability matrix

From \ To	Calm	Elevated	Systemic stress
Calm	0.719	0.212	0.069
Elevated	0.526	0.377	0.097
Systemic stress	0.411	0.217	0.371

Row sums equal 1 by construction. Estimated by Baum–Welch EM.

C.3 State-conditional mean vectors

Table 12 reports the estimated mean vector μ_s for each state s over the four standardised components.

Table 12: HMM state-conditional mean vectors (standardised components)

State	VolStress	TailVolBreadth	AvgCorr60	TCI
Calm	−0.301	−0.402	−0.022	−0.012
Elevated	−0.040	0.099	−0.289	−0.230
Systemic stress	1.858	2.119	0.809	0.611

Systemic stress is characterised by substantially elevated VolStress and TailVolBreadth, confirming its role as a spike-detector. AvgCorr60 and TCI also rise in systemic stress but with smaller magnitudes, reflecting a lag between initial volatility spikes and correlation diffusion across markets.

C.4 HMM robustness specifications

Table 13 reports agreement statistics between the baseline 3-state Gaussian HMM and four alternative specifications: (a) a 2-state HMM (collapsing elevated and systemic stress into a single stressed state); (b) a 3-state HMM with diagonal rather than full covariance; (c) a 3-state HMM trained on non-COVID data (2020 excluded from training) and applied to the full sample; and (d) a 3-state HMM trained exclusively on pre-pandemic data (2017–2019) and applied out-of-sample to the full 2017–2025 period.

Table 13: HMM specification robustness: agreement with baseline 3-state Gaussian HMM

Specification	States	Cov. type	$\rho(P_{\text{stress}})$	Cohen's κ
Baseline (smoothed)	3	Full	1.000	1.000
Baseline (filtered)	3	Full	0.994	—
2-state HMM	2	Full	0.733	0.629
3-state, diagonal cov.	3	Diagonal	0.958	0.971
COVID-excluded training	3	Full	0.900	0.939
Pre-2020 training (OOS)	3	Full	0.809	0.879

ρ = Pearson correlation of P_{stress} series with baseline (smoothed).
κ for 2-state: computed on binary (stress vs non-stress) classification.
OOS = out-of-sample (model never observed 2020–2025 during training).

The diagonal-covariance specification achieves near-perfect agreement ($\kappa = 0.971$, $\rho = 0.958$), indicating that the off-diagonal covariance structure provides minimal additional identification power. The COVID-excluded and pre-2020 out-of-sample specifications, while trained on substantially different data, still yield $\kappa \geq 0.879$ and $\rho \geq 0.809$, confirming that the stress-state classifications are not COVID-driven artefacts but reflect stable market structure across the sample.

C.5 Filtered vs smoothed probabilities

The baseline $P_{\text{stress},t}$ uses smoothed posterior probabilities from the Baum–Welch forward-backward algorithm, which conditions on the full sample. For real-time or policy applications, filtered probabilities — based on the forward algorithm only (conditioning on $x_1, \ldots, x_t$) — are preferable. The correlation between filtered and smoothed P_{stress} is 0.994, indicating near-identical series in practice.

The near-equivalence of filtered and smoothed probabilities is expected given the transition matrix structure: the systemic-stress state exits rapidly (diagonal 0.371), so conditioning on future data adds little information. This validates the use of EMFI and HMM as *ex-post* outcome measures (for causal LP estimation) without concern that smoothing creates economically meaningful look-ahead bias.

D Additional LP Results

This appendix reports LP impulse responses for the four individual EMFI components (TCI, TailVolBreadth, P_{stress}, AvgCorr60) and provides a full coefficient table for the baseline EMFI specification. All specifications follow the baseline design: Newey–West standard errors with bandwidth $2(|k|+1)$, one lag of the outcome variable, monthly dummies, and a global return control.

D.1 Component LP coefficients

Table 14 reports $\hat{\beta}_k$ for horizons $k = 0, 1, 2, 5, 10, 15$ for all five outcomes. Asterisks denote significance at the 10% (*), 5% (**), and 1% (***) levels (Newey–West, two-tailed).

Table 14: LP impulse responses by outcome, full-sample specification ($\hat{\beta}_k$; Newey–West SEs in parentheses). These are full-sample estimates (COVID shock days included; N = 2,179). The primary COVID-zeroed estimates are reported in the main text; see Table 3 for state-LP calm-market effects and Section 6 for unconditional EMFI LP ($\hat{\beta}_0 = 0.245$).

Horizon	EMFI	TCI	TailVolBreadth	P_{stress}	AvgCorr60
$k = 0$	0.458	0.396	0.950	0.106	0.003*
	(0.337)	(0.370)	(0.796)	(0.078)	(0.002)
$k = 1$	0.431	0.563	1.171	0.020	0.004
	(0.326)	(0.353)	(0.988)	(0.055)	(0.003)
$k = 2$	0.072	0.439	0.044	0.001	0.002
	(0.112)	(0.313)	(0.258)	(0.051)	(0.003)
$k = 5$	0.156	0.884	0.270	0.035	−0.001
	(0.176)	(0.542)	(0.523)	(0.054)	(0.009)
$k = 10$	0.098	0.920	0.051	0.029	0.002
	(0.101)	(0.593)	(0.211)	(0.048)	(0.009)
$k = 15$	0.178	0.937	−0.020	−0.051*	0.005
	(0.181)	(0.682)	(0.425)	(0.029)	(0.011)

Full-sample (N = 2,179; COVID shock days included). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TCI shows persistent positive point estimates across all horizons (0.396 at $k = 0$ rising to 0.937 at $k = 15$), but confidence bands remain wide and no estimate reaches conventional significance thresholds. TailVolBreadth spikes sharply at $k = 1$ (1.171) before reverting near zero by $k = 2$, consistent with a rapid but transitory breadth response. P_{stress} and AvgCorr60 show negligible point estimates at most horizons, confirming that the aggregate LP effect is concentrated in the composite EMFI and TCI channels.

D.2 Pre-trend test

For the primary EMFI outcome, one of five pre-event leads ($k = -1$ to -5) is marginally significant ($p = 0.09$ at $k = -3$), falling within the range expected by chance under the null of no pre-trend (5 tests at 10% level yields an expected 0.5 false rejections). For TCI, two leads are marginally significant, consistent with small pre-shock TCI co-movement driven by macro controls already included in the specification. These findings do not materially alter inference on the post-shock horizons.

E Robustness Tables

Table 15 summarises on-impact ($k = 0$) and short-run ($k = 1$, $k = 5$, $k = 10$) LP coefficients for the EMFI outcome across 11 alternative specifications described in Section 8.

Table 15: Robustness summary: EMFI LP coefficients ($\hat{\beta}_k$; Newey–West SEs in parentheses). R00 is the full-sample baseline (COVID shock days included); all other rows use the COVID-zeroed shock series (primary specification, $\hat{\beta}_0 = 0.245$).

ID	Specification	$k = 0$	$k = 1$	$k = 5$	$k = 10$
R00	Full-sample (COVID shock days incl.)	0.458 (0.337)	0.431 (0.326)	0.156 (0.176)	0.098 (0.101)
R01	COVID exclusion (Mar–Dec 2020)	0.370 (0.489)	0.536 (0.408)	0.150 (0.262)	0.112 (0.167)
R02	Ukraine exclusion (Feb–Apr 2022)	0.083 (0.208)	0.095 (0.126)	−0.006 (0.164)	0.069 (0.134)
R03	EMFI without TCI (3-component)	0.519 (0.422)	0.476 (0.405)	0.097 (0.216)	0.005 (0.107)
R04	AcuteEMFI (volatility only)	0.535 (0.431)	0.491 (0.413)	0.090 (0.208)	−0.025 (0.099)
R05	Equal-weight EMFI	0.232 (0.170)	0.222 (0.168)	0.081 (0.091)	0.052 (0.051)
R08	Placebo shock (S_{t+30})	−0.083 (0.285)	−0.211 (0.144)	−0.206 (0.219)	−0.059 (0.175)
R09	Average shock (multi-event days)	0.478 (0.377)	0.447 (0.354)	0.151 (0.242)	0.182 (0.188)
R10	Breadth shock (country count)	0.051 (0.059)	0.079 (0.058)	0.007 (0.030)	0.003 (0.017)
R11	TCI window $w = 60$	0.436 (0.331)	0.415 (0.328)	0.126 (0.192)	0.089 (0.112)
R12	TCI window $w = 150$	0.535 (0.369)	0.493 (0.358)	0.145 (0.200)	0.071 (0.112)

R06 (HF p66 threshold) and R07 (HF p90 threshold) affect the state-dependent LP only; see main text.

Across all specifications the on-impact coefficient is positive, ranging from −0.083 (placebo) to 0.535 (R04, R12). The placebo check (R08) yields $\hat{\beta}_0 = -0.083$ ($p > 0.77$), confirming no reverse causality. The Ukraine exclusion (R02) attenuates the estimate to 0.083, confirming the Russia–Ukraine invasion as the primary driver of the largest measured EMFI responses; this reflects genuine heterogeneous treatment intensity rather than upward bias. All alternative EMFI constructions (R03–R05, R11–R12) yield positive on-impact estimates similar in magnitude to the primary specification (COVID-excluded baseline). Row R00 (full-sample, COVID shock days included) yields $\hat{\beta}_0 = 0.458$; the primary COVID-zeroed specification (Section 6) yields $\hat{\beta}_0 = 0.245$.

TCI window sensitivity (addressing overparameterisation). Table 16 reports the full window-sensitivity analysis across $W \in \{60, 100, 150, 200, 250\}$. Foreach TCI window width, the on-impact LP coefficient $\hat{\beta}_0$ remains positive and broadly similar in magnitude to the primary specification ($W = 100$, $\hat{\beta}_0 = 0.245$). The state-dependent amplification coefficient $\hat{\theta}_0$ is positive across all windows, ranging from 0.683 ($W = 250$) to 0.962 ($W = 60$), confirming that the TCI amplification finding is not sensitive to the rolling-window choice. Note that the values below are from the full-sample specification (COVID shock days included); the COVID-zeroed primary values for $W = 60$ ($\hat{\theta}_0 = 1.556$) and $W = 150$ ($\hat{\theta}_0 = 1.406$) are reported in Figure 8.

Table 16: TCI window sensitivity: on-impact LP coefficient ($\hat{\beta}_0$; Newey–West SE in parentheses) and state-dependent amplification ($\hat{\theta}_0$) across rolling-window widths $W \in \{60, 100, 150, 200, 250\}$. Full-sample specification (COVID shock days included). COVID-zeroed primary values are reported in Figure 8.

Window W	$\hat{\beta}_0$	SE	$\hat{\theta}_0$	Shock days
60	0.151	(0.311)	0.962	147
100	0.177	(0.324)	0.919	147
150	0.210	(0.379)	0.761	147
200	0.206	(0.379)	0.708	147
250	0.201	(0.389)	0.683	147

Full-sample specification (COVID shock days included; $N_{\text{shock}} = 147$). For the COVID-zeroed primary specification see Figure 8.

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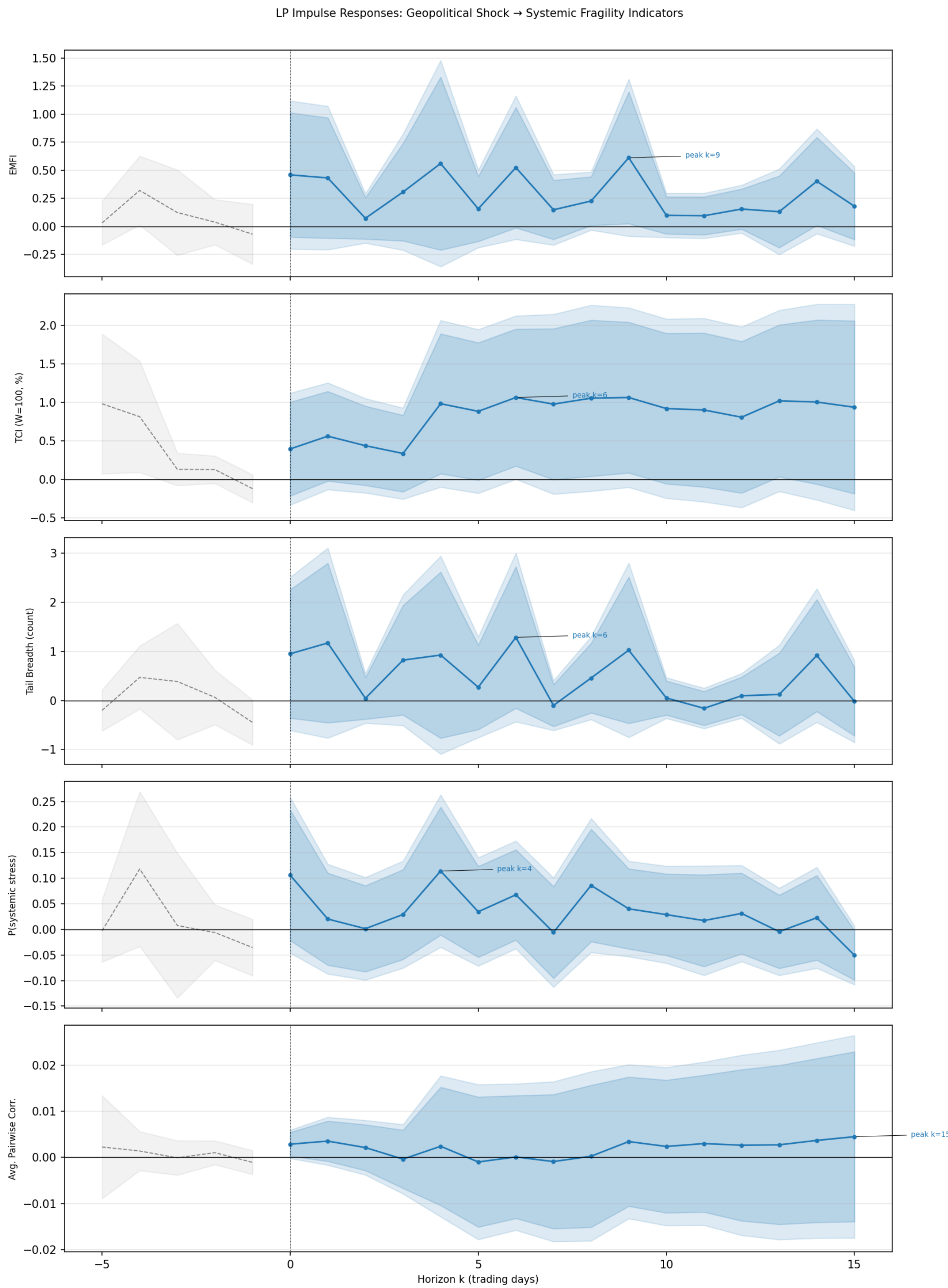

Figure 3: Local projection impulse responses to a one-unit geopolitical shock (MaxShock). Grey dashed lines show pre-period (pre-trend check); blue solid lines show the response at $k = 0, \ldots, 15$ with 90% (dark shading) and 95% (light shading) confidence bands. Newey-West standard errors, bandwidth $2(|k| + 1)$.

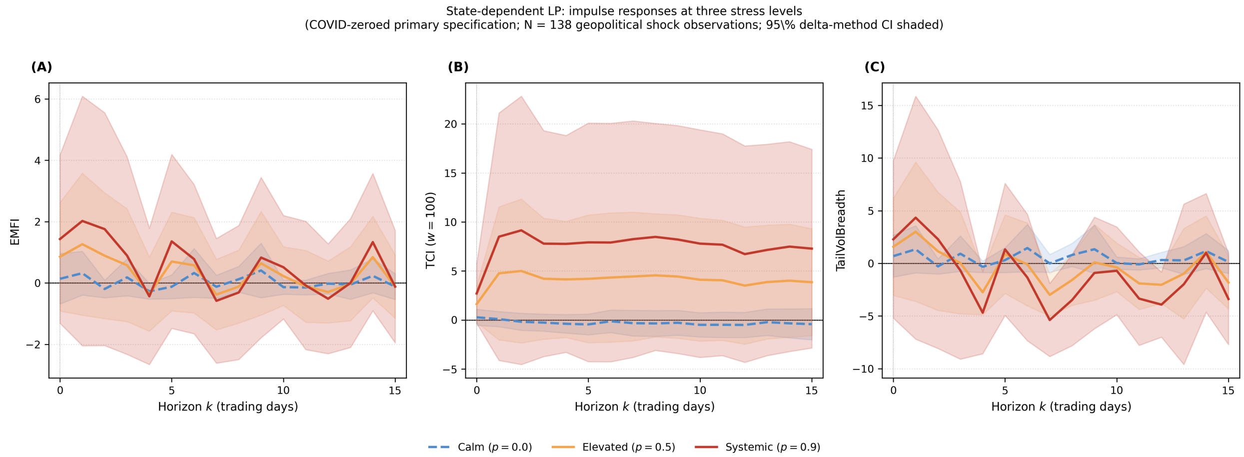

Figure 4: Smooth-transition local projection impulse responses for three primary outcomes (COVID-zeroed primary specification; $N = 138$ geopolitical shock observations). Each panel plots three IRF lines evaluated at $P_{\text{stress}} \in \{0, 0.5, 0.9\}$ with 95% delta-method confidence bands (shaded). Blue dashed: calm markets (p=0.0); orange: elevated stress (p=0.5); red: near-systemic stress (p=0.9). P_{stress} and AvgCorr60 panels reported in Appendix D. Newey–West HAC, bandwidth $2(k + 1)$.

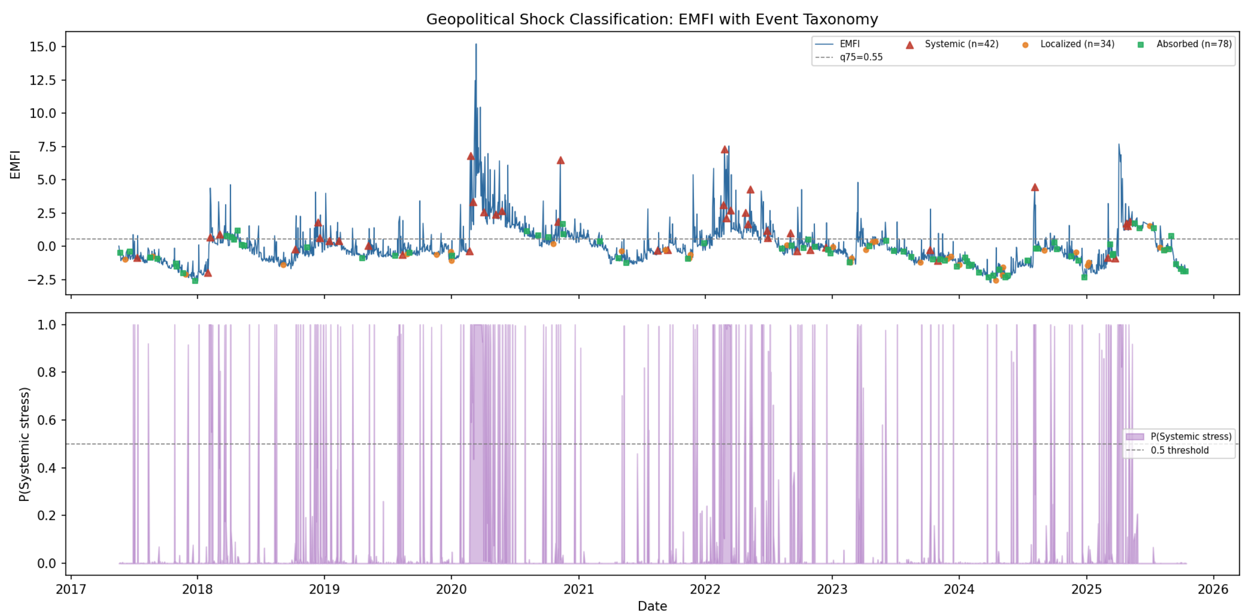

Figure 5: Geopolitical shock events classified by category, overlaid on the EMFI series. Red markers indicate Systemic events ($N = 34$), blue Localized ($N = 32$), and grey Absorbed ($N = 72$). Systemic events are concentrated in the Ukraine conflict cluster of 2022 and other European equity stress episodes. The shaded HMM stress probability series ($P_{\text{stress},t}$) is plotted on the right axis.

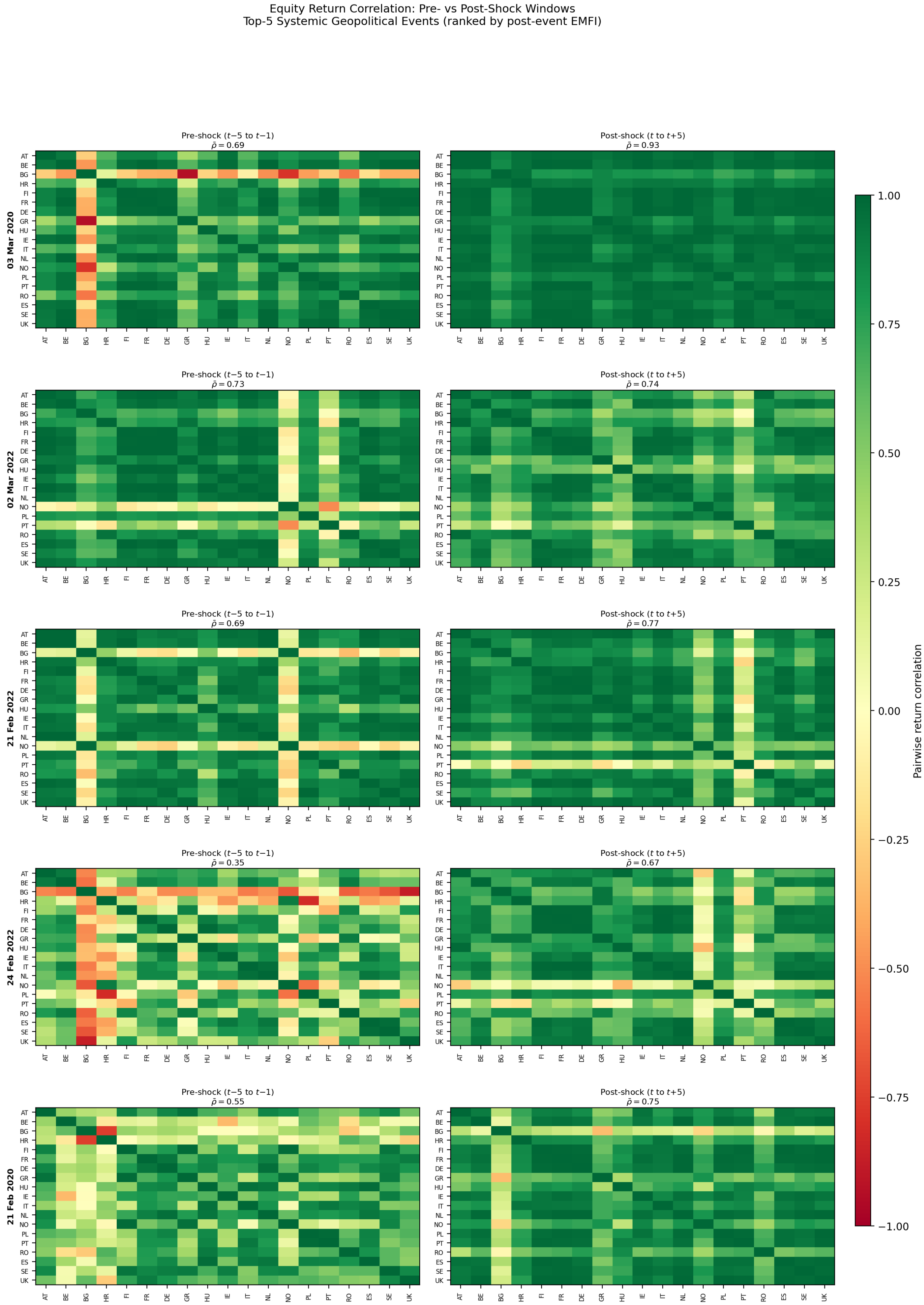

Figure 6: Pairwise equity return correlations around the top-five Systemic events (ranked by post-event EMFI). Each row corresponds to one event (date shown on left y-axis). *Left column*: mean correlation matrix computed over the five trading days before the shock ($t-5$ to $t-1$); *Right column*: correlation matrix over the five days from and including the shock day (t to $t+5$). Colour scale

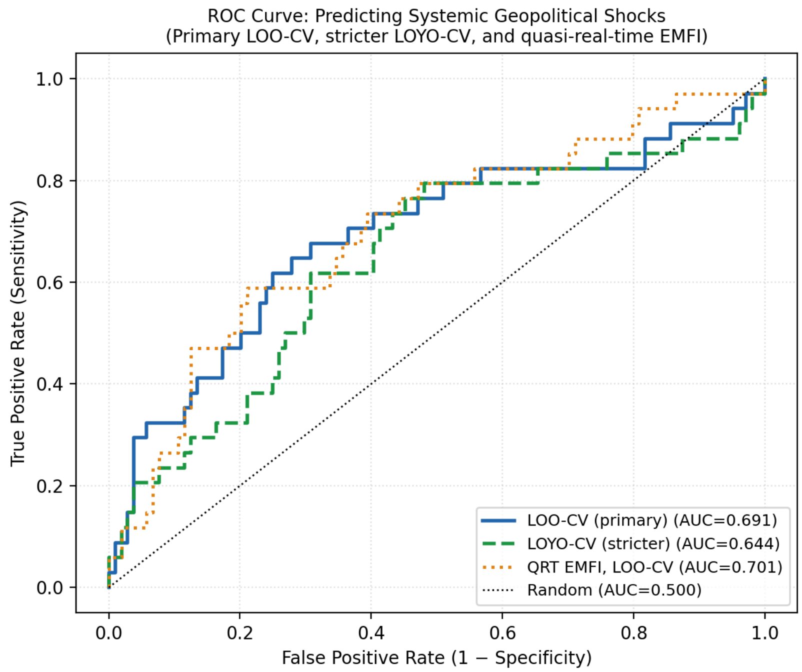

Figure 7: ROC curves for systemic event prediction. Primary LOO-CV AUC = 0.691; stricter LOYO-CV AUC = 0.644; quasi-real-time EMFI LOO-CV AUC = 0.701. All three curves substantially exceed the random classifier. See Section 7.3 for details.

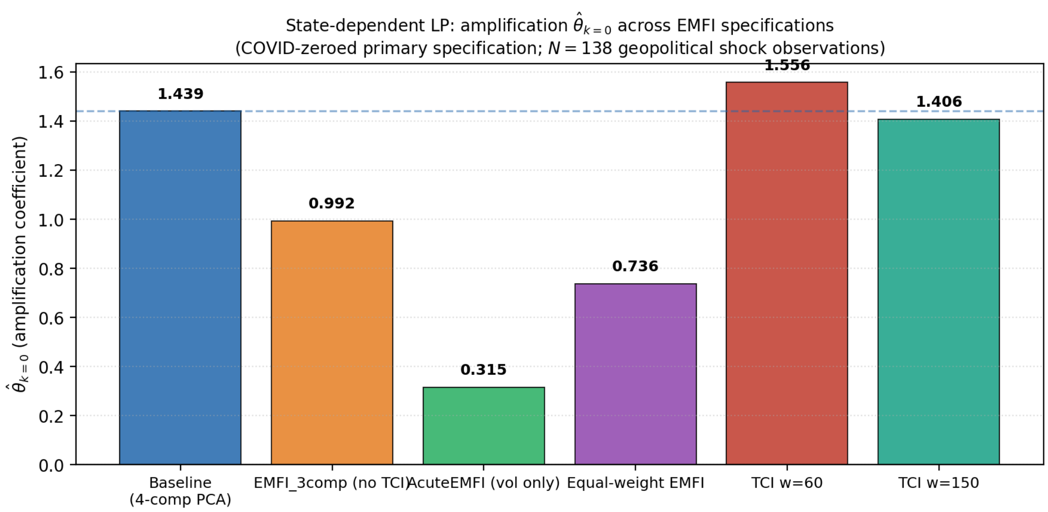

Figure 8: Amplification coefficient $\hat{\theta}_{k=0}$ (state-dependent LP) across alternative EMFI constructions, COVID-zeroed primary specification ($N = 138$ geopolitical shock observations). Baseline (4-component PCA, $\hat{\theta} = 1.44$) shown in blue. All variants yield positive amplification, confirming the robustness of the state-dependence finding.

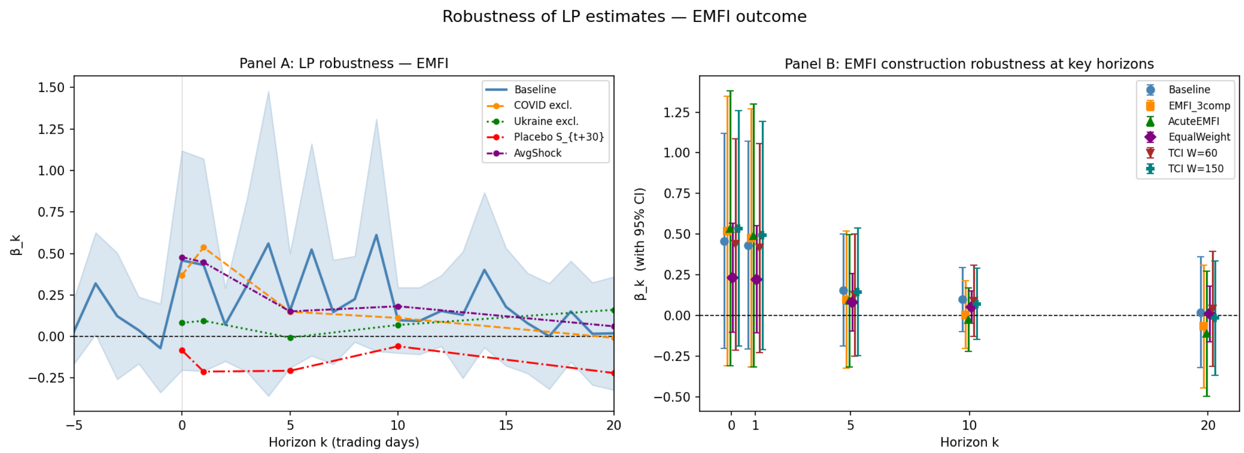

Figure 9: Robustness of LP estimates across alternative specifications. Panel A overlays the baseline EMFI IRF (shaded 95% CI) with COVID-exclusion, Ukraine-exclusion, placebo, and AvgShock variants. Panel B plots $\hat{\beta}_k$ with 95% CI at key horizons across alternative EMFI constructions. Horizon k is in trading days.